RAGBoost: Robust Tabular Learning via Retrieval-Augmented and Ancillary-Guided Gradient Boosting

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Abstract

Gradient Boosted Decision Trees (GBDTs) remain a leading approach for tabular learning, offering strong predictive performance, computational efficiency, and robustness to heterogeneous feature types. Yet their standard second-order leaf updates suffer from a node-level statistical limitation: as trees grow deeper, sparse leaves provide increasingly unreliable gradient estimates, while global regularization cannot adapt to local uncertainty or residual structure. We propose RAGBoost, a retrieval-augmented and ancillary-guided boosting framework for robust tabular learning. RAGBoost augments standard GBDT optimization with a two-layer node-adaptive correction mechanism. First, it retrieves historically observed leaves with similar gradient-state configurations to form node-conditional gradient priors, stabilizing variance-dominated sparse-leaf updates through empirical shrinkage. Second, it uses ancillary residual correction to detect and mitigate bias-dominated within-leaf residual structure that remains unresolved by the main tree. These corrections modify only leaf values, preserving the original tree structure and inference pipeline. Empirically, RAGBoost achieves the best overall average rank across 40 heterogeneous tabular benchmarks in our evaluation, while remaining strongly competitive with recent deep tabular and tabular foundation model baselines. On TabReD, an industrial-scale temporal-shift benchmark, RAGBoost achieves competitive performance against strong baselines, including XGBoost and the base TabM variant, suggesting improved robustness under realistic distribution drift.

1 Introduction

Tabular data, organized in row–column structures, underpin a vast range of scientific and industrial decision systems, from financial risk scoring and clinical diagnosis to recommendation engines and logistics scheduling. Among the learning paradigms developed for such data, Gradient Boosted Decision Trees (GBDTs) have maintained a dominant position over the past two decades [Friedman, 2001]. Implementations such as XGBoost, LightGBM, and CatBoost achieve strong predictive accuracy

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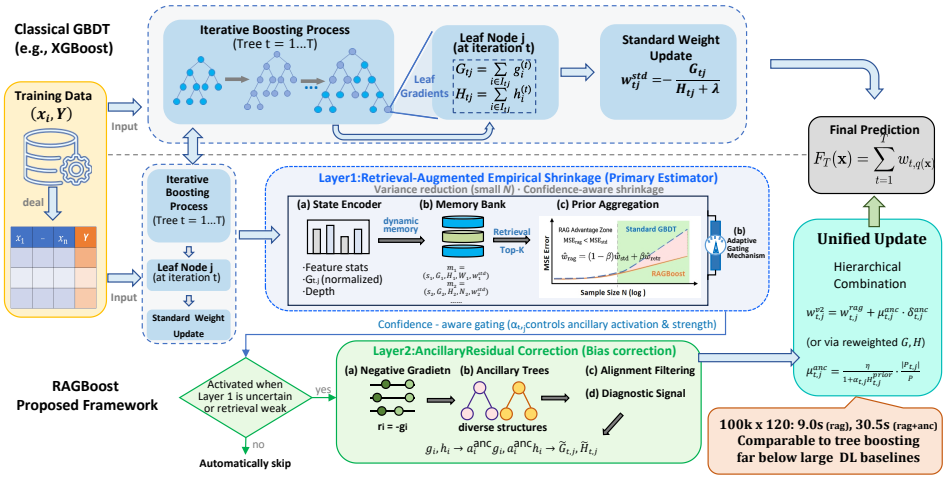


Figure 1: Overview of the RAGBoost Framework Architecture.

across heterogeneous feature types, tolerate missing values naturally, and remain computationally tractable at scale—properties that make them indispensable in practice [Chen and Guestrin, 2016, Ke et al., 2017, Prokhorenkova et al., 2018].

Yet a fundamental statistical tension lies at the heart of gradient boosting. As trees grow deeper, the number of training samples routed to each leaf node decreases exponentially. Under second-order optimization, the leaf weight update is computed from the sum of per-sample first- and second-order gradients; when this sum is taken over sparse leaf nodes (i.e., leaves receiving very few samples), the resulting estimate has high variance. Further deepening the tree may reduce approximation bias in local sub-regions, but exponentially decreases per-leaf sample counts, leading to high-variance weight estimates. This reflects a classical statistical principle: estimation from small samples is inherently unstable, while variance control typically relies on shrinkage that introduces bias [Stein, 1956, Efron and Morris, 1973]. Such a variance–bias tension is also widely observed in modern machine learning: global regularization stabilizes predictions by reducing variance, but operates uniformly across all nodes regardless of local sample size. Moreover, it may oversmooth local sub-regions with distinct distributions, treating valid signals as noise and leading to systematic estimation bias in these areas [Zhang and Yang, 2021].

Existing approaches—whether stronger regularization within GBDT or architectural redesigns in deep learning—share a key limitation: they do not leverage information from previously observed leaf nodes. The boosting process remains memoryless across iterations, and each leaf independently estimates its update using only its local samples. However, structurally similar gradient configurations often recur during training. This suggests a missing mechanism: allowing sparse or heterogeneous leaves to condition their updates on historical gradient statistics from similar leaves. Analogous ideas have proven effective in language modeling via retrieval-augmented generation [Lewis et al., 2020], motivating a retrieval-based perspective for gradient boosting.

We introduce RAGBoost, a two-layer framework that addresses both variance- and bias-dominated regimes. Layer 1 (retrieval-augmented empirical shrinkage) stabilizes sparse leaves by retrieving structurally similar historical leaves and forming a node-conditional gradient-space prior. Layer 2 (ancillary residual correction) addresses unresolved local bias by extracting leaf-specific correction signals from auxiliary trees trained under diverse inductive biases. Adaptive gating balances the two layers based on sample size and retrieval reliability. The resulting estimator is node-adaptive, data-driven, and grounded in empirical Bayes principles, with provable asymptotic recovery to the standard GBDT update when sufficient local data is available.

Our contributions are as follows:

- We identify the high-variance leaf estimation problem in deep GBDTs as a node-level statistical estimation problem, and formally characterize how second-order leaf updates become increasingly unstable as tree depth reduces the effective sample size per leaf.
- We propose RAGBoost, a retrieval-augmented boosting framework that augments standard second-order GBDT optimization with a two-layer node-adaptive correction mechanism: retrieval-based empirical shrinkage injects node-conditional gradient priors for sparse leaves, while ancillary residual correction detects and mitigates unresolved within-leaf bias, without altering tree structure.
- We show that RAGBoost provides intrinsic interpretability signals beyond standard feature importance, including leaf correction states, retrieval provenance traces, retrieval reliance scores, ancillary residual diagnostics, and source-decomposed SHAP-compatible attributions.
- Across 40 heterogeneous tabular benchmarks, RAGBoost achieves a leading overall rank, rivaling recent foundation and deep tabular models, while retaining GBDT-style simplicity and efficiency. On the industrial-scale TabReD benchmark, it achieves better overall performance than XGBoost and the base TabM, demonstrating robustness under distribution drift.

2 Related Work

2.1 Small-Sample Estimation and Shrinkage

The variance problem in sparse leaf nodes is an instance of a classical challenge in statistics: estimating the mean of a distribution from a small number of observations yields estimates with high variance, and naive plug-in estimators are inadmissible under squared error loss. The James–Stein estimator [Stein, 1956, James et al., 1961] provided a foundational shrinkage correction that trades a small bias for a large reduction in variance. Subsequent developments in empirical Bayes estimation [Robbins, 1992, Efron and Morris, 1975, Efron, 2016] formalized the idea of using the marginal distribution of observed data to construct adaptive priors, enabling estimators that borrow strength across related estimation problems—a principle recently extended to high-dimensional machine learning settings where shrinkage intensity is estimated from data rather than fixed in advance [Plessis et al., 2023].

In the context of gradient boosting, each leaf node update is precisely a small-sample mean estimation problem: the leaf weight is proportional to the sum of first-order gradients divided by the sum of second-order gradients, and when only a few samples reach a given leaf, this ratio has high variance. Standard GBDT regularization adds a fixed penalty λ to the denominator—equivalent to assuming a fixed prior strength regardless of local sample size—whereas empirical Bayes shrinkage adapts regularization intensity to each node's sample size and estimation uncertainty. Existing GBDT regularization therefore lacks the adaptive, evidence-calibrated shrinkage that small-sample estimation theory prescribes.

2.2 Optimization and Regularization in Gradient Boosting

GBDTs perform functional gradient descent in the space of tree ensembles, with each tree fit to second-order pseudo-residuals [Friedman, 2001]. XGBoost [Chen and Guestrin, 2016] introduced a regularized objective that penalizes leaf weights and tree complexity; LightGBM [Ke et al., 2017] proposed histogram-based splitting and gradient-based one-side sampling to accelerate training; CatBoost [Prokhorenkova et al., 2018] addressed target leakage through ordered boosting for categorical features. These advances collectively address runtime efficiency and certain types of overfitting, but their regularization mechanisms share a common structure: global, tree-level controls fixed prior to or independent of local node statistics.

More targeted approaches include minimum child weight constraints, feature subsampling, and gradient clipping. These reduce variance on average but do so by discarding or constraining data rather than improving estimation quality where data is inherently limited. Crucially, none of these mechanisms uses information from previously observed leaf nodes—the optimization process remains memoryless across boosting iterations.

2.3 Deep and Retrieval-Augmented Models for Tabular Data

Recent years have seen growing interest in deep learning and foundation models for tabular data. Representative approaches include Transformer-based architectures such as TabTransformer [Huang et al., 2020] and FT-Transformer [Gorishniy et al., 2021], sequential attention in TabNet [Arik and Pfister, 2020], lightweight ensembling in TabM [Gorishniy et al., 2025], and foundation-style models such as TabPFN [Hollmann et al., 2025] and iLTM [Bonet et al., 2025]. In parallel, retrieval-augmented learning has become an effective paradigm in language modeling [Lewis et al., 2020, Gao et al., 2023], and has recently been explored for tabular prediction. For example, TabR [Gorishniy et al., 2023] retrieves similar training instances in embedding space and incorporates their labels into prediction, resembling a learned form of kNN-based smoothing.

Despite their strong empirical performance, these methods primarily improve tabular learning through architectural design, large-scale pretraining, representation learning, or prediction-stage retrieval. They do not directly target the second-order optimization dynamics of GBDTs. In contrast, RAGBoost uses retrieval inside the boosting optimization process: the memory bank stores historical leaf-level gradient statistics, and retrieved evidence is used to correct local leaf-weight updates through a quality-aware gate. This makes RAGBoost structurally different from representation-level retrieval methods and complementary to deep tabular architectures. It refines the statistical estimation step of boosted trees rather than replacing GBDTs with a new model family.

3 Methodology

In this section, we present RAGBoost, a retrieval-augmented and ancillary-guided framework for improving GBDTs through node-adaptive leaf-value correction. The method introduces a two-layer correction mechanism that stabilizes sparse-leaf estimation and resolves within-leaf residual structure, while preserving the original tree structure and inference pipeline.

3.1 Problem Setup

At boosting iteration t, a regression tree partitions the input space into leaf regions $\{R_{t,j}\}_{j=1}^{L}$. For leaf j, let $I_{t,j} = \{i : x_i \in R_{t,j}\}$ and $N_{t,j} = |I_{t,j}|$. The standard second-order GBDT leaf weight is

$$w_{t,j}^{\mathrm{std}} = -\frac{G_{t,j}}{H_{t,j} + \lambda}, \qquad G_{t,j} = \sum_{i \in I_{t,j}} g_i^{(t)}, \quad H_{t,j} = \sum_{i \in I_{t,j}} h_i^{(t)}. \tag{1}$$

The instability of this estimator comes from its local sample dependence. Conditional on the current model, leaf assignment, and Hessians, if gradients are uncorrelated with bounded variance and $h_i^{(t)} \geq h_{\min} > 0$, then

$$\mathrm{Var}(w_{t,j}^{\mathrm{std}} \mid \cdot) \leq \frac{\sigma_j^2}{N_{t,j} h_{\min}^2}. \tag{2}$$

Thus sparse leaves suffer from high-variance Newton updates, while heterogeneous leaves may still contain unresolved residual structure even when $N_{t,j}$ is moderate. RAGBoost addresses these two regimes through a two-layer correction: retrieval-augmented empirical shrinkage and ancillary residual correction.

3.2 Layer 1: Retrieval-Augmented Empirical Shrinkage

For each leaf j, RAGBoost constructs a state vector $s_{t,j} \in \mathbb{R}^q$ from structural statistics, normalized gradient statistics, and node depth. A memory bank $\mathcal{M}$ stores historical high-confidence leaf entries

$$m_{t,j} = (s_{t,j},\, G_{t,j},\, H_{t,j},\, N_{t,j},\, w_{t,j}^{\mathrm{std}}).$$

To avoid temporal or evaluation leakage, retrieval at iteration t uses only training-set memory entries constructed from previous boosting iterations; current-tree leaves are inserted into $\mathcal{M}$ only after all leaf weights of tree t are finalized.

For an admitted leaf, RAGBoost retrieves the top-K nearest memory entries by distance $\rho_{jk} = \|s_{t,j} - s^{(k)}\|_2$, with weights

$$\pi_{jk} := \frac{\exp(\kappa/(1+\rho_{jk}))}{\sum_{k'=1}^{K} \exp(\kappa/(1+\rho_{jk'}))}.$$

The retrieved statistics define gradient-space priors

$$G_{t,j}^{\text{prior}} = \sum_k \pi_{jk} G^{(k)}, \qquad H_{t,j}^{\text{prior}} = \sum_k \pi_{jk} H^{(k)}.$$

The Layer 1 update is

$$w_{t,j}^{\text{rag}} := -\frac{G_{t,j} + \alpha_{t,j} G_{t,j}^{\text{prior}}}{H_{t,j} + \alpha_{t,j} H_{t,j}^{\text{prior}} + \lambda}, \tag{3}$$

where the gate $\alpha_{t,j}$ is defined as

$$\alpha_{t,j} := \frac{c}{N_{t,j}^{\gamma}(\tilde{\zeta}_{t,j}^{\text{retr}} + \tilde{\Delta}_{t,j}^2 + \epsilon)}. \tag{4}$$

Here $\tilde{\zeta}_{t,j}^{\text{retr}}$ measures retrieved-weight dispersion and $\tilde{\Delta}_{t,j}^2$ measures geometric drift. Therefore, retrieval is strongest for sparse leaves with reliable neighbors and is suppressed when the retrieved prior is uncertain or distant. If a leaf is not admitted, RAGBoost sets $w_{t,j}^{\text{rag}} = w_{t,j}^{\text{std}}$. Full admission criteria and normalization details are given in Appendix A.2.

This update is equivalent to an empirical shrinkage estimator. Specifically, defining

$$\hat{w}_{\text{retr}} := -\frac{G_{t,j}^{\text{prior}}}{H_{t,j}^{\text{prior}} + \lambda},$$

we have

$$w_{t,j}^{\text{rag}} := a_{t,j} w_{t,j}^{\text{std}} + b_{t,j} \hat{w}_{\text{retr}},$$

where $a_{t,j}$ decreases and $b_{t,j}$ increases with $\alpha_{t,j}$. Under small regularization, this reduces to an approximate convex combination of the standard estimator and a retrieval-based node-conditional prior. The proof is provided in Appendix A.2.

3.3 Layer 2: Ancillary Residual Correction

Layer 1 stabilizes sparse leaves but cannot fully resolve leaves containing local residual substructure unmatched by the memory bank. RAGBoost therefore fits ancillary trees $\{f_t^{(p)}\}_{p=1}^{P}$ to the current negative gradients $r_i^{(t)} = -g_i^{(t)}$. These trees are used only as transient residual probes: they do not enter the ensemble, are not retained at inference time, and only affect the main tree through leaf-weight computation.

For main leaf j, the alignment between the residual vector $\mathbf{r}_{t,j}$ and ancillary prediction vector $\hat{\mathbf{r}}_{t,j}^{(p)}$ is

$$A_{t,j}^{(p)} := \frac{\langle \mathbf{r}_{t,j}, \hat{\mathbf{r}}_{t,j}^{(p)} \rangle}{\|\mathbf{r}_{t,j}\|_2 \|\hat{\mathbf{r}}_{t,j}^{(p)}\|_2 + \epsilon}. \tag{5}$$

Ancillary trees satisfying $A_{t,j}^{(p)} > \tau$ form the valid set $\mathcal{P}_{t,j}$. In the scalar correction form, the ancillary signal is

$$\delta_{t,j}^{\text{anc}} := \sum_{p \in \mathcal{P}_{t,j}} \omega_{t,j}^{(p)} \bar{r}_{t,j}^{(p)}, \qquad \omega_{t,j}^{(p)} := \frac{A_{t,j}^{(p)}}{\sum_{p' \in \mathcal{P}_{t,j}} A_{t,j}^{(p')}}, \tag{6}$$

where $\bar{r}_{t,j}^{(p)}$ denotes the leaf-local prediction of ancillary tree p on leaf j. The final scalar update is

$$w_{t,j}^{\text{v2}} = w_{t,j}^{\text{rag}} + \mu_{t,j}^{\text{anc}} \delta_{t,j}^{\text{anc}}, \qquad \mu_{t,j}^{\text{anc}} = \frac{\eta}{1 + \alpha_{t,j} H_{t,j}^{\text{prior}}} \cdot \frac{|\mathcal{P}_{t,j}|}{P}. \tag{7}$$

Algorithm 1 RAGBoost iteration

Compute $g_i^{(t)}, h_i^{(t)}$ from the current model F_{t-1}.
Fit the main tree structure and obtain leaves $\{R_{t,j}\}_{j=1}^{L}$.
for each leaf $j = 1, \ldots, L$ **do**
    Compute $G_{t,j}, H_{t,j}, N_{t,j}$ and $w_{t,j}^{\mathrm{std}}$.
    Apply the retrieval admission criterion: for admitted leaves, retrieve previous memory entries and compute $w_{t,j}^{\mathrm{rag}}$; otherwise set $w_{t,j}^{\mathrm{rag}} = w_{t,j}^{\mathrm{std}}$.
end for
Fit P transient ancillary trees on $r_i^{(t)} = -g_i^{(t)}$ and extract valid ancillary signals.
for each leaf $j = 1, \ldots, L$ **do**
    Compute $w_{t,j}^{\mathrm{v2}}$ by scalar correction or diagnostic reweighting.
end for
Update $F_t = F_{t-1} + \nu f_t^{\mathrm{v2}}$.
Insert sufficiently sampled current-tree leaves into $\mathcal{M}$.

The factor $(1 + \alpha_{t,j} H_{t,j}^{\mathrm{prior}})^{-1}$ makes Layer 2 complementary to Layer 1: ancillary correction is reduced when retrieval is confident and becomes more active when retrieval is weak.

We also implement a diagnostic reweighting variant in which accepted ancillary trees induce sample-level Newton reweighting before computing the final leaf weight. This variant preserves the same main tree, retrieval prior, and safe fallback structure, and is accepted only through a local gain gate. Details are deferred to Appendix A.4.5–A.4.7.

3.4 Algorithm and Computational Complexity

Algorithm 1 summarizes one training iteration of RAGBoost. The procedure follows the standard GBDT pipeline for fitting the main tree structure, but replaces the standard leaf-weight estimation step with the proposed two-layer correction. Retrieval is performed only for admitted leaves using historical training-set memory entries from previous boosting iterations. Ancillary trees are fitted only as transient residual probes and are discarded after the current iteration.

The algorithm enforces a strict temporal order: memory entries from the current tree are inserted only after all its leaf weights have been finalized, preventing temporal or evaluation leakage. Both the memory bank and ancillary trees are used only during training and do not introduce additional inference-time components.

The additional training cost comes from approximate nearest-neighbor retrieval over admitted leaves and from fitting P transient ancillary trees. Using the notation of Appendix C, the overall training complexity is

$$\mathcal{O}\left(T\mathcal{C}_{\mathrm{tree}}(n,d) + TP\mathcal{C}_{\mathrm{anc}}(n,d) + TPn + \sum_{t=1}^{T} S_t \, T_{\mathrm{ANN}}(M_t, q, K) \right), \tag{8}$$

where S_t is the number of leaves admitted to retrieval at iteration t, and $T_{\mathrm{ANN}}(M_t, q, K)$ is the approximate-nearest-neighbor query cost for one admitted leaf. Detailed runtime decomposition and empirical time-cost analysis are provided in Appendix C. At inference time, RAGBoost has the same asymptotic prediction cost as a standard GBDT model with the same number and depth of trees.

4 Experiments

4.1 Experimental Setup

Benchmarks. We evaluate RAGBoost on two groups: (1) 40 public tabular datasets [Bischl et al., 2017, Gijsbers et al., 2019] (20 classification, 20 regression) with diverse scales and distributions, split 70/15/15; and (2) TabReD [Rubachev et al., 2024], an industrial benchmark of eight datasets featuring temporal distribution shifts, evaluated using official time-based splits.

Baselines. Comparisons include tree-based (XGBoost, CatBoost, LightGBM, RandomForest), neural (FT-Transformer), and recent tabular methods (TabPFN, TabR, TabM). On TabReD, we also include

Table 1: Aggregate performance on 40 public tabular datasets.

Task	Metric	RAGBoost	TabPFN	TabM	XGBoost	CatBoost	FT-Trans	TabR	LightGBM	RF
Classif.	Avg. Rank ↓	**3.2250**	4.5000	4.4750	4.9250	5.3000	5.8500	5.0750	5.5750	6.0750
	Avg. Rel. Score ↑	**0.6863**	0.6283	0.5511	0.4906	0.5085	0.4663	0.5022	0.5562	0.3435
	#Best ↑	**6**	7	2	0	0	1	4	1	0
Regress.	Avg. Rank ↓	**2.7250**	3.5250	4.3250	5.4000	5.2500	4.9500	5.6750	6.0500	7.1000
	Avg. Rel. Score ↑	**0.6917**	0.6856	0.5933	0.4444	0.4801	0.4831	0.4247	0.3779	0.2572
	#Best ↑	**7**	7	4	0	2	1	0	0	0
Overall	Avg. Rank ↓	**2.9750**	4.0125	4.4000	5.1625	5.2750	5.4000	5.3750	5.8125	6.5875

the tabm-mini-piecewiselinear variant, denoted as TabM†mini, following the official benchmark protocol.

Protocol and Metrics. Public datasets are tuned via Bayesian optimization (10 seeds); TabReD follows its official pipeline. We report Accuracy/RMSE, average rank, relative scores, and win counts for public data, while adhering to TabReD's official ROC AUC/RMSE and ranking metrics. Full implementation details are in Appendix D.

4.2 Main Results on 40 Public Datasets

Table 1 summarizes the aggregate performance of RAGBoost across 40 public tabular datasets. RAGBoost achieves the best overall average rank of **2.9750**, compared with 4.0125 for TabPFN, 4.4000 for TabM, and 5.1625 for XGBoost. It also achieves the top or tied-top performance on 13 of the 40 datasets and ranks among the top three on 27 datasets. These results demonstrate its strong aggregate performance across diverse classification and regression tasks.

RAGBoost shows consistent gains across both classification and regression. On classification tasks, TabPFN achieves one more dataset-level win than RAGBoost (7 vs. 6), but RAGBoost attains a better average rank and a higher average relative score. This suggests that although TabPFN can be highly competitive on selected datasets, RAGBoost provides stronger average-case robustness across the full classification suite.

The advantage is more pronounced on regression tasks. RAGBoost obtains the best average rank and the highest average relative score, and it ties with TabPFN for the largest number of wins (7 each). This pattern is consistent with the motivation of RAGBoost: regression losses directly expose the quality of leaf-value estimation, and high-variance Newton updates in sparse leaves can lead to large prediction errors. By improving the leaf-weight estimator rather than replacing the tree ensemble architecture, RAGBoost provides a direct correction to this failure mode.

4.3 Results on TabReD

We further evaluate RAGBoost on TabReD to test whether its gains persist under more realistic deployment conditions. Unlike the randomly split public benchmark, TabReD uses temporal splits, where test samples are drawn from later time periods than training samples. This setting is more challenging because it evaluates robustness under distribution shift rather than only in-distribution generalization.

Table 2 reports results on the eight TabReD tasks, where RAGBoost achieves an average rank of $\mathbf{3.56 \pm 2.21}$, securing the second position among all reported methods. It outperforms standard tree ensembles, MLP, FT-Transformer, TabR, and the base TabM model, with only $\text{TabM}^{\dagger}_{\text{mini}}$ achieving a slightly better rank due to its strong performance on specific regression tasks like Sberbank Housing and Maps Routing. Demonstrating robust performance across both classification and regression benchmarks, particularly under distribution shift, RAGBoost achieves a better average rank than the conventional GBDT baselines across the eight TabReD tasks, indicating that the proposed training-time correction enhances tree-based models beyond conventional i.i.d. settings. On classification tasks, it attains the best performance on HomeCredit Default and remains competitive overall, although representation-driven neural models (e.g., MLP, TabM) retain advantages in certain temporally shifted industrial scenarios such as Ecom Offers. Similarly, in regression, RAGBoost achieves the best or tied-best results on Delivery ETA and Weather while staying competitive elsewhere. These results

Table 2: Results on TabReD. Best, second-best, and third-best results are highlighted in **bold**, *italic*, and underline, respectively.

Methods	Classification (ROC AUC ↑)			Regression (RMSE ↓)					Avg. Rank
	Homesite	Ecom	Credit	Sberbank	Cooking	Delivery	Maps	Weather	
RAGBoost(ours)	0.9608	0.5844	**0.8677**	0.2440	0.4823	**0.5465**	0.1622	**1.4624**	*3.56*
XGBoost	0.9601	0.5763	*0.8670*	*0.2419*	0.4823	0.5468	0.1616	1.4671	4.44
LightGBM	0.9603	0.5758	0.8664	0.2468	0.4826	0.5468	0.1618	*1.4625*	5.19
CatBoost	0.9606	0.5596	0.8621	0.2482	0.4823	**0.5465**	0.1619	1.4688	5.69
RandomForest	0.9570	0.5764	0.8269	0.2640	0.4884	0.5959	0.1653	1.5838	9.25
MLP	0.9500	**0.6015**	0.8545	0.2508	0.4820	0.5504	0.1622	1.5470	6.38
FT-Transformer	0.9622	0.5775	0.8571	0.2440	0.4820	0.5542	0.1625	1.5104	6.00
TabR	0.9487	0.5943	0.8501	0.2820	0.4828	0.5514	0.1639	1.4666	7.50
TabM	**0.9647**	*0.5950*	0.8614	0.2467	**0.4802**	0.5502	*0.1612*	1.4706	3.69
$\text{TabM}^{\dagger}_{\text{mini}}$	*0.9631*	0.5930	0.8640	**0.2319**	**0.4802**	0.5516	**0.1609**	1.4672	**3.31**

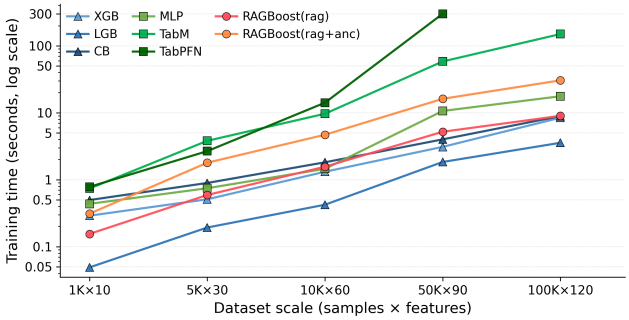


Figure 2: Empirical runtime comparison. RAGBoost(rag) denotes the retrieval-only variant, while RAGBoost(rag+anc) denotes the full model with ancillary correction. TabPFN's architecture imposes a hard limit of 50K samples, so the $100K \times 120$ setting exceeds its capacity. Full numerical results and theoretical complexity analysis are provided in Appendix C.

highlight that RAGBoost improves the competitiveness of boosted tree ensembles under temporal distribution shift while preserving their deployment efficiency.

4.4 Efficiency Analysis

We finally evaluate the computational cost of RAGBoost. A detailed theoretical complexity analysis is provided in Appendix C. Figure 2 reports the empirical training runtime across five synthetic data scales. We evaluate RAGBoost(rag) using the largest retrieval configuration selected in our benchmark experiments, with memory capacity set to 2,000 and Top-K=20, which provides a conservative estimate of retrieval overhead. Under this setting, RAGBoost(rag) introduces only moderate overhead: at 100000×120, it requires 9.003 ± 0.077 seconds, close to XGBoost (8.439 ± 0.087) and CatBoost (8.910 ± 0.009). The additional RAM usage remains within approximately 2–10 MB over the corresponding XGBoost baseline. These results indicate that the retrieval layer remains practically lightweight even under the largest selected retrieval configuration.

5 Mechanistic Analysis

We now analyze when and where RAGBoost's correction mechanism takes effect. To isolate distinct local estimation regimes, we synthesize five DAG-generated regression benchmarks, each designed to expose a specific failure mode of deep trees: sparse local regions, within-leaf residual heterogeneity, repeated local mechanisms, temporal distribution shift, and noisy rare subpopulations. Full data-generating process definitions are in Appendix E. All datasets share the same sample scale, feature dimension, and split protocol. We compare XGBoost, LightGBM, and five RAGBoost variants using RMSE.

Table 3: RMSE on five DAG-generated regression benchmarks. R denotes retrieval-augmented correction, A denotes ancillary correction, RandomMem replaces retrieved memory with random neighbor statistics, and NoGate removes dynamic gating.

Method	Sparse	Hetero	Repeated	Shift	NoisyRare	Avg.
XGBoost	1.154	1.257	1.141	1.284	1.492	1.266
LightGBM	1.162	1.244	1.118	1.292	1.465	1.256
RAGBoost-A	1.121	1.254	1.136	1.286	1.421	1.244
RAGBoost-R	1.143	1.217	1.122	1.288	1.356	1.225
RAGBoost-R-RandomMem	1.152	1.264	1.121	1.296	1.456	1.258
RAGBoost	**1.091**	**1.251**	**1.034**	**1.286**	**1.318**	**1.196**
RAGBoost-R+A-NoGate	1.291	1.262	1.253	1.314	1.322	1.288

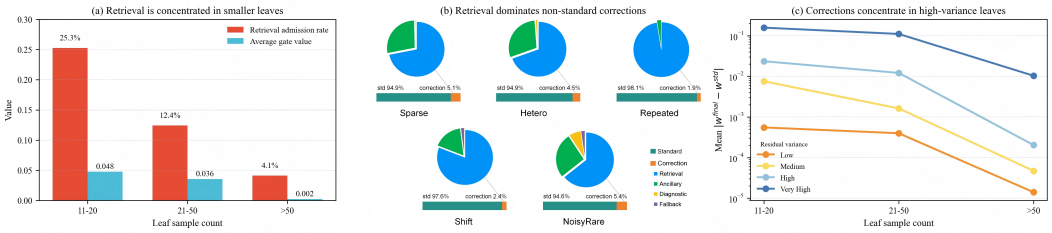


Figure 3: Where RAGBoost applies correction. (a) Retrieval admission rate and average gate value on DAG-Sparse across leaf-count buckets. (b) Sample-weighted proportions of standard and non-standard correction states on the five DAG benchmarks, visualized as five pie charts. (c) Mean correction magnitude by leaf sample count and residual variance. Together, the three panels show that RAGBoost is conservative overall, but applies stronger corrections in leaves that are smaller and exhibit higher residual uncertainty.

5.1 Which Component Matters?

We first ask whether the two correction layers of RAGBoost contribute differently under different data-generating regimes. Table 3 reports RMSE on the five DAG-generated benchmarks.

Table 3 shows that the two layers contribute in different regimes and that their gains depend on selective correction. Retrieval is the stronger component: RAGBoost-R improves the average RMSE from 1.256/1.266 for LightGBM/XGBoost to 1.225, with clear gains on Hetero and NoisyRare. The full model further reduces the average RMSE to 1.196, especially on Sparse (1.091) and Repeated (1.034), where sparse updates and recurring local mechanisms make historical gradient-state evidence useful. Ancillary correction alone is more modest, reaching 1.244 on average, consistent with its role as a transient residual probe. The controls further show that correction evidence must be gated: RandomMem remains close to base GBDT baselines (1.258 average), while removing the gate gives the worst average RMSE (1.288). Thus, the gate acts as a safeguard that suppresses unreliable retrieval, reduces ancillary correction when retrieval is confident, and falls back toward the standard leaf value when neither source is reliable.

5.2 Where Does RAGBoost Correct?

These diagnostics complement the ablation results by showing where the correction mechanism is activated. While Table 3 evaluates which components improve prediction, Figure 3 reveals how those components are distributed across leaves and datasets.

Figure 3 shows (a) retrieval admission rate and average gate value on DAG-Sparse grouped by leaf sample count; since the base learner uses a minimum child size of 20, only non-empty buckets are shown. Retrieval is clearly concentrated in smaller leaves: admission rate decreases from 25.3% for leaves with 11–20 samples, to 12.4% for leaves with 21–50 samples, and to 4.1% for leaves with more than 50 samples. The average gate value follows the same trend, dropping from 0.0478 to 0.0357 to 0.0024, which indicates that Layer-1 retrieval correction is mainly activated where the standard leaf estimate is more sample-limited. (b) The sample-weighted correction-state proportions

across the five DAG benchmarks show that standard updates remain dominant (94.6–98.1% of the sample-weighted correction-state mass), so RAGBoost does not globally perturb boosting. Among non-standard states, Retrieval is consistently the main correction mode, Ancillary provides a smaller secondary contribution, and Diagnostic and Fallback states are negligible; overall, the model relies primarily on gated retrieval with limited ancillary refinement. (c) Correction magnitude is highly localized: the largest deviations occur in leaves with high residual variance, especially when leaf count is small. For example, mean correction magnitude reaches 0.1580 in leaves with 11–20 samples and very high residual variance, and 0.1106 in leaves with 21–50 samples under the same variance condition, whereas it is close to zero in large low-variance leaves, confirming that corrections are targeted adjustments, not a global offset, applied to leaves whose standard estimates are statistically weak or structurally unresolved.

5.3 Interpretability Diagnostics

A further advantage of RAGBoost is that its interpretability is built into the optimization procedure rather than attached after training. Standard GBDT models typically expose only aggregate split- or gain-based feature importance, which does not reveal why a particular leaf update was trusted, corrected, or suppressed. In contrast, RAGBoost records the diagnostic pathway of each leaf. Every leaf update can be classified as a standard update, retrieval-augmented update, ancillary-corrected update, diagnostic-reweighted update, or fallback update. This converts the correction mechanism into an explicit audit signal: the model can distinguish leaves that are locally stable from leaves that require historical borrowing or residual-structure correction.

These diagnostics further extend to sample-level and evidence-level explanations. Because each sample inherits the state of the leaves it visits across boosting iterations, RAGBoost can identify samples that repeatedly require non-standard correction. Moreover, the retrieval module provides provenance traces showing which historical leaves support a prediction, while ancillary trees expose residual structure that remains unresolved by the main tree. Thus, RAGBoost offers a layered explanation interface: state-level correction labels, retrieval reliance, ancillary residual diagnostics, and compatibility with standard SHAP analysis through the final scalar leaf values. Appendix B provides the complete interpretability framework, including formal state definitions, consistency scores, state-conditional feature importance, source-decomposed SHAP values, and retrieval provenance.

6 Conclusion

This paper presented RAGBoost, a retrieval-augmented and ancillary-guided framework that improves gradient boosted decision trees through node-adaptive leaf-value correction. RAGBoost addresses two limitations of standard second-order boosting: high-variance updates in sparse leaves and unresolved residual structure in heterogeneous leaves. Through retrieval-augmented empirical shrinkage and ancillary residual correction, the method stabilizes local leaf estimation without changing the tree structure or increasing inference-time complexity. Experiments on 40 public tabular datasets and the TabReD temporal-shift benchmark show that RAGBoost achieves competitive overall performance against classical tree ensembles, deep tabular models, and recent tabular methods, while the diagnostic results indicate that its corrections are conservative and localized. Nevertheless, RAGBoost remains within the gradient boosting paradigm and may not fully capture the smooth high-order representations learned by alternative model families. Future work may therefore investigate hybrid designs that combine node-adaptive boosting corrections with representation learning or cross-paradigm ensembles, aiming to further improve predictive performance while preserving the efficiency and interpretability required for large-scale tabular learning.

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A Complete Framework, Proofs, and Supplementary Material

Below is a detailed definition of the notations used in this paper.

Boosting and tree structure:

Symbol	Definition
t	Boosting iteration index
j	Leaf node index within a tree
L	Number of leaves per tree
$R_{t,j}$	Leaf region of node j at iteration t
$I_{t,j}$	Sample index set $\{i : x_i \in R_{t,j}\}$
$N_{t,j}$	Leaf sample count $\lvert I_{t,j}\rvert$
$F_t(x)$	Ensemble prediction after t iterations

Gradient statistics:

Symbol	Definition
$g_i^{(t)},\ h_i^{(t)}$	First- and second-order loss gradients at sample i
$G_{t,j},\ H_{t,j}$	Aggregated gradients and Hessians over $I_{t,j}$
$r_i^{(t)}$	Negative gradient (pseudo-residual): $r_i^{(t)} = -g_i^{(t)}$
σ_j^2	Upper bound on $\mathrm{Var}(g_i^{(t)} \mid \cdot)$ within leaf j
$h_{\min}$	Lower bound on Hessians: $h_i^{(t)} \geq h_{\min} > 0$

Leaf weight estimators:

Symbol	Definition
$w_{t,j}^{\text{std}}$	Standard leaf weight: $-G_{t,j}/(H_{t,j} + \lambda)$
$w_{t,j}^{\text{rag}}$	Retrieval-augmented (Layer 1) leaf weight
$w_{t,j}^{\text{v2}}$	Final RAGBoost leaf weight
$w_{t,j}^{*}$	Population-optimal leaf weight (Definition A.1, Section A.6)
$w_{t,j}^{\dagger}$	Conditional Newton step: $-\mathbb{E}[G_{t,j} \mid \cdot]/(H_{t,j} + \lambda)$
$\hat{w}_{\text{retr}}$	Retrieval-prior estimator: $-G_{t,j}^{\text{prior}}/(H_{t,j}^{\text{prior}} + \lambda)$
λ	Fixed L2 regularization penalty

Memory and retrieval (Layer 1):

Symbol	Definition
$s_{t,j} \in \mathbb{R}^q$	State vector of leaf node j at iteration t
$\mathcal{M}$	Memory bank of historical leaf entries
$m_{t,j}$	Memory entry: $(s_{t,j}, G_{t,j}, H_{t,j}, N_{t,j}, w_{t,j}^{\text{std}})$
K	Number of retrieved nearest neighbors
ρ_{jk}	Euclidean distance $\lVert s_{t,j} - s^{(k)}\rVert_2$
κ	Softmax temperature for retrieval weights
π_{jk}	Retrieval weight for neighbor k
$G_{t,j}^{\text{prior}},\ H_{t,j}^{\text{prior}}$	Aggregated gradient-space priors

Gating (Layer 1):

Symbol	Definition
$\alpha_{t,j}$	Adaptive gating coefficient for the retrieval-based, node-conditional prior
$\zeta_{t,j}^{\text{retr}}$	Retrieved neighbor weight variance: $\mathrm{Var}_{k\sim\pi}(w^{(k)})$
$\tilde{\zeta}_{t,j}^{\text{retr}}$	Normalized retrieval variance: $\zeta_{t,j}^{\text{retr}}/\bar{\zeta}_t$
$\hat{\Delta}_{t,j}^2$	Geometric drift: $\sum_k \pi_{jk}\|s^{(k)} - s_{t,j}\|_2^2$
$\tilde{\Delta}_{t,j}^2$	Normalized geometric drift: $\hat{\Delta}_{t,j}^2/\bar{\Delta}_t^2$
c, γ, ϵ	Gating hyperparameters
$\tilde{\beta}_{t,j}$	Approximate shrinkage coefficient (Proposition 3.1)

Ancillary correction (Layer 2):

Symbol	Definition
P	Number of ancillary trees
$f_t^{(p)}$	Ancillary tree p at iteration t
$\hat{r}_i^{(p)}$	Ancillary tree p's leaf prediction for sample i
$A_{t,j}^{(p)}$	Directional alignment score (cosine similarity)
τ	Alignment threshold
$\mathcal{P}_{t,j}$	Valid ancillary set: $\{p : A_{t,j}^{(p)} > \tau\}$, possibly further filtered by diagnostic criteria
$\omega_{t,j}^{(p)}$	Alignment- or diagnostic-proportional weight for ancillary tree p
$\bar{r}_{t,j}^{(p)}$	Mean ancillary prediction within leaf j
$\delta_{t,j}^{\text{anc}}$	Ancillary correction term
$\mu_{t,j}^{\text{anc}}$	Ancillary gating factor
η	Global scaling constant for $\mu_{t,j}^{\text{anc}}$

Diagnostic reweighting refinement (Layer 2, compatible implementation):

Symbol	Definition
$V_{t,j}$	Within-leaf residual variance: $\mathrm{Var}_{i\in I_{t,j}}(r_i^{(t)})$
$E_{t,j}^{(p)}$	Residual-structure explanation score of ancillary tree p inside leaf j
τ_E	Explanation-score threshold for accepting diagnostic ancillary information
$s_i^{(p)}$	Sample-level responsibility score induced by ancillary tree p
s_i^{anc}	Aggregated sample-level ancillary responsibility score
a_i^{anc}	Ancillary-induced Newton reweighting factor for sample i
λ_{anc}	Strength of diagnostic reweighting
$s_{\max}$	Upper clipping constant for diagnostic responsibility
$\tilde{g}_i^{(t)},\ \tilde{h}_i^{(t)}$	Ancillary-reweighted gradient and Hessian
$\tilde{G}_{t,j},\ \tilde{H}_{t,j}$	Aggregated ancillary-reweighted gradient and Hessian over $I_{t,j}$

MSE analysis:

Symbol	Definition
σ_{retr}^2	Variance of the retrieval-prior estimator
Δ	Retrieval bias: $\mathbb{E}[\hat{w}_{\text{retr}}] - w_{t,j}^{\dagger}$
σ_{rag}^2	Variance of $w_{t,j}^{\text{rag}}$
σ_{anc}^2	Variance of $\delta_{t,j}^{\text{anc}}$
Δ_{anc}	Compound correction bias (Theorem 3.2)
N_{rag}^*	Sample-count threshold for Theorem 3.1

A.1 Problem Formalization and Leaf Weight Variance

A.1.1 Setup

Consider a gradient boosting model [Friedman, 2001] at iteration t, producing a regression tree that partitions the input space into L disjoint leaf regions $\{R_{t,j}\}_{j=1}^{L}$. For leaf node j, the sample index set is $I_{t,j} = \{i : x_i \in R_{t,j}\}$ with cardinality $N_{t,j} = |I_{t,j}|$. The standard second-order leaf weight update [Chen and Guestrin, 2016] is

$$w_{t,j}^{\text{std}} = -\frac{G_{t,j}}{H_{t,j} + \lambda}, \qquad G_{t,j} = \sum_{i \in I_{t,j}} g_i^{(t)}, \quad H_{t,j} = \sum_{i \in I_{t,j}} h_i^{(t)},$$

where $g_i^{(t)} = \partial \ell(y_i, F_{t-1}(x_i))/\partial F_{t-1}(x_i)$ and $h_i^{(t)} = \partial^2 \ell(y_i, F_{t-1}(x_i))/\partial F_{t-1}(x_i)^2$ are the first- and second-order loss gradients at sample i, and $\lambda > 0$ is a fixed regularization penalty.

We will measure the quality of leaf weight estimates against two targets: the population-optimal leaf weight $w_{t,j}^*$ and the conditional Newton step $w_{t,j}^\dagger$, both formally defined in Section A.6 (Definition A.1). Informally, $w_{t,j}^*$ is the probability limit of $w_{t,j}^{\text{std}}$ as the leaf sample count grows, and $w_{t,j}^\dagger$ is the conditional expectation of the standard weight given the leaf assignment and Hessians; under mild conditions, $w_{t,j}^\dagger \to w_{t,j}^*$ as $\lambda/N_{t,j} \to 0$.

A.1.2 Leaf Weight Variance Bound (Lemma 3.1)

Lemma 3.1 (Leaf Weight Variance Bound)**.** *Conditional on F_{t-1}, the leaf assignment $I_{t,j}$, and the Hessians $\{h_i^{(t)}\}_{i \in I_{t,j}}$, assume the gradients $\{g_i^{(t)}\}_{i \in I_{t,j}}$ are uncorrelated with $\text{Var}(g_i^{(t)} \mid \cdot) \le \sigma_j^2$, and that $h_i^{(t)} \ge h_{\min} > 0$. Then*

$$\text{Var}\Big(w_{t,j}^{std} \mid I_{t,j}, \{h_i^{(t)}\}\Big) \le \frac{\sigma_j^2}{N_{t,j}\, h_{\min}^2}.$$

Proof. All statements below are conditional on F_{t-1}, $I_{t,j}$, and $\{h_i^{(t)}\}_{i \in I_{t,j}}$. Under this conditioning, the denominator $D_{t,j} := H_{t,j} + \lambda$ is deterministic, and all randomness in $w_{t,j}^{\text{std}} = -G_{t,j}/D_{t,j}$ resides in the numerator $G_{t,j}$.

Step 1 (Numerator variance). By the uncorrelatedness assumption on $\{g_i^{(t)}\}_{i \in I_{t,j}}$:

$$\text{Var}(G_{t,j} \mid \cdot) = \sum_{i \in I_{t,j}} \text{Var}(g_i^{(t)} \mid \cdot) \le N_{t,j}\, \sigma_j^2.$$

Remark on the uncorrelatedness assumption. Conditional on F_{t-1} and the leaf assignment $I_{t,j}$, each gradient $g_i^{(t)} = \partial \ell(y_i, F_{t-1}(x_i))/\partial F_{t-1}(x_i)$ depends only on the pair (x_i, y_i) and the deterministic function F_{t-1}. Under i.i.d. data, residual noise across distinct samples is therefore uncorrelated given F_{t-1} and $I_{t,j}$.

Step 2 (Denominator lower bound). From $h_i^{(t)} \ge h_{\min}$:

$$D_{t,j} = H_{t,j} + \lambda \ge N_{t,j}\, h_{\min} + \lambda \ge N_{t,j}\, h_{\min}.$$

Step 3 (Variance propagation). Since $w_{t,j}^{\text{std}} = -G_{t,j}/D_{t,j}$ with $D_{t,j}$ deterministic:

$$\text{Var}(w_{t,j}^{\text{std}} \mid \cdot) = \frac{\text{Var}(G_{t,j} \mid \cdot)}{D_{t,j}^2} \le \frac{N_{t,j}\, \sigma_j^2}{(N_{t,j}\, h_{\min})^2} = \frac{\sigma_j^2}{N_{t,j}\, h_{\min}^2}. \qquad \square$$

A.1.3 Failure Regime Analysis

The bound in Lemma 3.1 reveals a structural instability: variance decays as $\mathcal{O}(1/N_{t,j})$, so deep trees that route few samples to a leaf incur high-variance weight estimates regardless of λ. Since λ is a global constant, standard regularization cannot adapt to node-specific sample counts, motivating a node-conditional, data-adaptive mechanism.

Following the classical bias–variance decomposition [Geman et al., 1992], we decompose the conditional MSE of $w_{t,j}^{\text{std}}$ around $w_{t,j}^{\dagger}$ (Definition A.1, Section A.6) and identify two failure regimes:

- *Regime I (Variance-dominated).* $\text{Var}(w_{t,j}^{\text{std}}) \gg \text{Bias}^2(w_{t,j}^{\text{std}})$. By Lemma 3.1, this occurs when $N_{t,j}$ is small, i.e., for sparse leaf nodes (leaves receiving very few samples). This is the classical setting in which shrinkage estimators reduce MSE [James et al., 1961, Efron and Morris, 1973, Xing et al., 2021, Li and Ignatiadis, 2025].
- *Regime II (Bias-dominated).* The leaf lies in a local sub-region with distinct distributions that the greedy splitting trajectory [Friedman, 2001] fails to isolate; approximation bias dominates, and historical retrieval alone is insufficient. In this regime, the main tree may route samples with different residual structures into the same leaf, so a single leaf weight averages over incompatible local directions.

RAGBoost addresses both regimes through a two-layer correction: a retrieval-augmented empirical shrinkage (Layer 1, Section A.2) and an ancillary residual correction (Layer 2, Section A.4). The Layer 2 mechanism is presented in its original scalar correction form and, as a compatible refinement, in a diagnostic reweighting form that allows ancillary trees to influence the Newton statistics used by the main leaf update.

A.2 Layer 1: Retrieval-Augmented Empirical Shrinkage

Layer 1 performs node-conditional empirical shrinkage: it stabilizes a noisy leaf estimate by borrowing strength from historically similar leaves—a principle rooted in the empirical Bayes tradition [Efron and Morris, 1973], where information from related estimation problems is pooled to improve individual estimates.

A.2.1 State Representation and Memory Bank

For leaf node j at iteration t, we form a state vector $s_{t,j} \in \mathbb{R}^q$ concatenating three types of context:

- Structural statistics: feature means and quantiles computed over the sample set $I_{t,j}$;
- Gradient statistics: the normalized mean gradient $\bar{g}_{t,j} = G_{t,j}/N_{t,j}$;
- Depth: the depth of node j in tree t.

A memory bank $\mathcal{M}$ dynamically accumulates entries from high-confidence leaf nodes (those with $N_{t,j} \geq N_{\min}$). Each entry is

$$m_{t,j} = \left(s_{t,j},\ G_{t,j},\ H_{t,j},\ N_{t,j},\ w_{t,j}^{\text{std}}\right).$$

Entries are appended only after all leaf weights of tree t have been finalized, so no entry from the current tree is visible during its own construction; it becomes retrievable only from iteration $t+1$ onward. To keep storage bounded, $\mathcal{M}$ is capped at capacity C; once $|\mathcal{M}| = C$, the oldest entry is discarded under FIFO order. Hence $|\mathcal{M}| \leq C$ for the entire run, yielding a bounded memory footprint without clustering or compression.

Before encoding the retrieval state, continuous input features are standardized using training-set statistics; structural statistics are therefore computed on standardized features. The gradient term uses the leaf-average $\bar{g}_{t,j} = G_{t,j}/N_{t,j}$ rather than the aggregated gradient to remove leaf-size dependence, and the depth coordinate is normalized by the maximum tree depth.

A.2.2 Retrieval and Prior Aggregation

Given a query state $s_{t,j}$, we retrieve the top-K nearest neighbors $\{m^{(k)}\}_{k=1}^{K}$ from $\mathcal{M}$ by Euclidean distance $\rho_{jk} = \|s_{t,j} - s^{(k)}\|_2$. Aggregation weights are computed via a softmax kernel with temperature $\kappa > 0$:

$$\pi_{jk} = \frac{\exp(\kappa/(1+\rho_{jk}))}{\sum_{k'=1}^{K} \exp(\kappa/(1+\rho_{jk'}))}.$$

This retrieval-and-weight scheme is a locality-sensitive adaptation of kernel-weighted local estimation [Nadaraya, 1964, Watson, 1964]. The retrieved neighbors aggregate into gradient-space priors:

$$G_{t,j}^{\text{prior}} = \sum_{k=1}^{K} \pi_{jk}\, G^{(k)}, \qquad H_{t,j}^{\text{prior}} = \sum_{k=1}^{K} \pi_{jk}\, H^{(k)}.$$

In practice, this top-K retrieval is implemented with approximate nearest-neighbor (ANN) search, which maintains retrieval accuracy while scaling efficiently to large memory banks and large-scale data regimes; see Appendix C for the corresponding cost term $T_{\text{ANN}}(M_t, q, K)$.

A.2.3 Retrieval-Augmented Weight and Adaptive Gating

The Layer 1 leaf weight is

$$w_{t,j}^{\text{rag}} = -\frac{G_{t,j} + \alpha_{t,j}\, G_{t,j}^{\text{prior}}}{H_{t,j} + \alpha_{t,j}\, H_{t,j}^{\text{prior}} + \lambda},$$

with the adaptive gating coefficient

$$\alpha_{t,j} = \frac{c}{N_{t,j}^{\gamma}\left(\tilde{\zeta}_{t,j}^{\text{retr}} + \tilde{\Delta}_{t,j}^{2} + \epsilon\right)},$$

where $\tilde{\zeta}_{t,j}^{\text{retr}}$ and $\tilde{\Delta}_{t,j}^{2}$ are normalized versions of the retrieved-neighbor weight variance and geometric drift, respectively (normalization defined in Section A.2.5 below), and $c, \gamma, \epsilon > 0$ are hyperparameters.

The gating coefficient satisfies two key properties:

1. Sample-size awareness: $\alpha_{t,j} \to 0$ as $N_{t,j} \to \infty$, so $w_{t,j}^{\text{rag}} \to w_{t,j}^{\text{std}}$, recovering the standard update asymptotically.
2. Retrieval uncertainty suppression: large $\tilde{\zeta}_{t,j}^{\text{retr}}$ or $\tilde{\Delta}_{t,j}^{2}$ reduces $\alpha_{t,j}$, preventing noisy or geometrically distant priors from dominating.

A.2.4 Shrinkage Equivalence (Proposition 3.1)

Proposition 3.1 (Shrinkage Equivalence)**.** *Define the retrieval-prior estimator* $\hat{w}_{retr} := -G_{t,j}^{prior}/(H_{t,j}^{prior} + \lambda)$*. Then*

$$w_{t,j}^{rag} = a_{t,j}\, w_{t,j}^{std} + b_{t,j}\, \hat{w}_{retr},$$

with $a_{t,j} > 0$ *strictly decreasing in* $\alpha_{t,j}$ *and* $b_{t,j} > 0$ *strictly increasing in* $\alpha_{t,j}$*. Under* $\lambda \ll \min(H_{t,j}, H_{t,j}^{prior})$*, this reduces to an approximate convex combination* $w_{t,j}^{rag} \approx (1 - \tilde{\beta}_{t,j})\, w_{t,j}^{std} + \tilde{\beta}_{t,j}\, \hat{w}_{retr}$ *with* $\tilde{\beta}_{t,j} \in (0, 1)$*.*

Proof. Exact affine decomposition. Let $D = H_{t,j} + \alpha_{t,j}\, H_{t,j}^{\text{prior}} + \lambda$. Define

$$a_{t,j} = \frac{H_{t,j} + \lambda}{D}, \qquad b_{t,j} = \frac{\alpha_{t,j}(H_{t,j}^{\text{prior}} + \lambda)}{D}.$$

Direct substitution gives

$$a_{t,j}\, w_{t,j}^{\text{std}} + b_{t,j}\, \hat{w}_{\text{retr}} = -\frac{G_{t,j}}{D} - \frac{\alpha_{t,j}\, G_{t,j}^{\text{prior}}}{D} = -\frac{G_{t,j} + \alpha_{t,j}\, G_{t,j}^{\text{prior}}}{D} = w_{t,j}^{\text{rag}}.$$

Monotonicity. Since $\partial D/\partial \alpha_{t,j} = H_{t,j}^{\text{prior}} > 0$, the denominator D is strictly increasing in $\alpha_{t,j}$. The numerator of $a_{t,j}$ is constant, so $a_{t,j}$ is strictly decreasing. For $b_{t,j}$, differentiation yields $\partial b_{t,j}/\partial \alpha_{t,j} = (H_{t,j}^{\text{prior}} + \lambda)(H_{t,j} + \lambda)/D^2 > 0$, confirming strict increase.

Approximate convex combination. Define $\eta_{t,j} := a_{t,j} + b_{t,j}$ and $\tilde{\beta}_{t,j} := b_{t,j}/\eta_{t,j}$. Then

$$\eta_{t,j} = 1 + \frac{\alpha_{t,j}\, \lambda}{D}, \qquad \tilde{\beta}_{t,j} = \frac{\alpha_{t,j}(H_{t,j}^{\text{prior}} + \lambda)}{(H_{t,j} + \lambda) + \alpha_{t,j}(H_{t,j}^{\text{prior}} + \lambda)} \in (0, 1),$$

and $w^{\text{rag}}_{t,j} = \eta_{t,j}\Big[(1-\tilde{\beta}_{t,j})\, w^{\text{std}}_{t,j} + \tilde{\beta}_{t,j}\, \hat{w}_{\text{retr}}\Big]$. Under $\lambda \ll \min\{H_{t,j}, H^{\text{prior}}_{t,j}\}$ with $\alpha_{t,j} \le \alpha_{\max}$, we have $|\eta_{t,j} - 1| \le \alpha_{\max}\lambda / \min\{H_{t,j}, H^{\text{prior}}_{t,j}\} = \mathcal{O}(\lambda/H)$, so $\eta_{t,j} \approx 1$ and the expression reduces to the approximate convex combination. □

This structure is formally analogous to a James–Stein-type shrinkage estimator [James et al., 1961], with the key difference that the shrinkage target is a retrieval-based, node-conditional prior in gradient space that adapts to each leaf's structural context, rather than a fixed global mean.

A.2.5 Normalization of the Gating Coefficient

The raw quantities $\zeta^{\text{retr}}_{t,j} = \text{Var}_{k\sim\pi}(w^{(k)})$ and $\hat{\Delta}^2_{t,j} = \sum_k \pi_{jk}\|s^{(k)} - s_{t,j}\|_2^2$ have different physical dimensions. To ensure dimensional consistency of $\alpha_{t,j}$, we normalize each by its running mean over all leaves accumulated up to iteration t:

$$\tilde{\zeta}^{\text{retr}}_{t,j} = \frac{\zeta^{\text{retr}}_{t,j}}{\bar{\zeta}_t}, \qquad \tilde{\Delta}^2_{t,j} = \frac{\hat{\Delta}^2_{t,j}}{\bar{\Delta}^2_t},$$

where $\bar{\zeta}_t = (1/|\mathcal{L}_t|)\sum_{(t',j')\in\mathcal{L}_t} \zeta^{\text{retr}}_{t',j'}$ and $\bar{\Delta}^2_t = (1/|\mathcal{L}_t|)\sum_{(t',j')\in\mathcal{L}_t} \hat{\Delta}^2_{t',j'}$ are computed over the set $\mathcal{L}_t$ of all leaf indices processed up to iteration t. With this normalization, both quantities become dimensionless relative reliability indicators. Their additive combination is not intended to imply identical distributions or variances; instead, it provides a conservative penalty that decreases retrieval strength when either uncertainty source becomes large.

A.2.6 Admission Criterion: When to Activate Retrieval

The retrieval mechanism and the continuous gate $\alpha_{t,j}$ (Sections A.2.2–A.2.3) are designed to operate only when a leaf is genuinely difficult to estimate. Activating retrieval indiscriminately would waste computation on well-behaved leaves and risk introducing unnecessary bias. We therefore define a discrete admission criterion that acts as a pre-retrieval filter.

Definition (Admission Criterion)**.** *A leaf node j at iteration t is admitted to the retrieval pipeline if and only if*

$$\text{admit}(j) = \underbrace{\mathbb{1}\Big[N_{t,j} < \rho \cdot \tilde{N}_t\Big]}_{\textit{sparsity}} \;\vee\; \underbrace{\mathbb{1}\Big[V_{t,j} > \theta_{var} \cdot \tilde{V}_t \;\wedge\; N_{t,j} \ge N^{het}_{\min}\Big]}_{\textit{local heterogeneity}}, \tag{A.1}$$

where:

- $N_{t,j} = |I_{t,j}|$ *is the leaf sample count;*
- $V_{t,j} = \text{Var}_{i\in I_{t,j}}(r^{(t)}_i)$ *is the within-leaf variance of the pseudo-residuals* $r^{(t)}_i = -g^{(t)}_i$*;*
- $\tilde{N}_t = \text{median}_j\, N_{t,j}$ *and* $\tilde{V}_t = \text{median}_{j:V_{t,j}>0}\, V_{t,j}$ *are tree-level medians used for adaptive thresholding;*
- $\rho \in (0,1)$*,* $\theta_{var} > 1$*, and* $N^{het}_{\min} \in \mathbb{N}$ *are hyperparameters. Default values used in all experiments are* $\rho = 0.5$*,* $\theta_{var} = 1.5$*, and* $N^{het}_{\min} = 20$*.*

If $\text{admit}(j) = 0$, the leaf bypasses retrieval entirely, taking the standard weight $w^{\text{std}}_{t,j}$ without computing $G^{\text{prior}}_{t,j}$ or $H^{\text{prior}}_{t,j}$, and $\alpha_{t,j}$ is effectively zero.

Rationale and Connection to Theory.

- Sparsity criterion ($N_{t,j} < \rho\tilde{N}_t$). This directly targets the variance-dominated regime characterized by Lemma 3.1 and Theorem 3.1. Because $\text{Var}(w^{\text{std}}_{t,j}) = \mathcal{O}(1/N_{t,j})$, leaves with sample counts substantially below the tree median are precisely those for which Theorem 3.1 guarantees $\text{MSE}(w^{\text{rag}}_{t,j}) < \text{MSE}(w^{\text{std}}_{t,j})$. Using a relative threshold (fraction of median) auto-calibrates the criterion to the dataset scale.

- Heterogeneity criterion ($V_{t,j} > \theta_{\text{var}} \tilde{V}_t$). This identifies leaves where the main tree has not locally converged: the residual variance remains high, signaling that the greedy splitting strategy has failed to isolate the local sub-region with distinct distributions. In such leaves, retrieval may reduce MSE even when $N_{t,j}$ is moderate. The guard $N_{t,j} \geq N_{\min}^{\text{het}}$ is statistically necessary because the sample variance estimator is unreliable for small N (its own relative variance scales as $\mathcal{O}(1/N)$), preventing spurious admission of small leaves where high observed variance is merely sampling noise.

Relationship to the Continuous Gate $\alpha_{t,j}$. The admission criterion and the continuous gating coefficient are complementary, not redundant:

- $\text{admit}(j)$ is a pre-retrieval discrete filter based solely on local leaf statistics, determining whether retrieval computation is warranted at all.
- $\alpha_{t,j}$ is a post-retrieval continuous modulator that adjusts the shrinkage strength based on retrieval-side signals ($\tilde{\zeta}_{t,j}^{\text{retr}}$, $\tilde{\Delta}_{t,j}^2$), defined only when $\text{admit}(j) = 1$.

Together they implement a two-level safeguard: the admission criterion prevents unwarranted activation on easy leaves, while $\alpha_{t,j}$ prevents unreliable retrieved priors from dominating on admitted leaves. Rejected leaves retain $w_{t,j}^{\text{std}}$ exactly, matching the safe fallback behavior established in Proposition 3.2(ii).

Asymptotic Consistency. As $N_{t,j} \to \infty$ for all leaves (e.g., as the dataset size grows while tree depth remains bounded), the sparsity criterion eventually fails globally, and the heterogeneity criterion becomes increasingly unlikely to fire because residual variance decays. Hence the admission rate vanishes asymptotically, and all leaves revert to $w_{t,j}^{\text{std}}$, which is consistent with the asymptotic recovery property proved in Proposition 3.2(i).

A.3 Proof of Theorem 3.1 (Conditional MSE Reduction of Layer 1 Retrieval Correction)

We now establish that Layer 1 reduces MSE in the variance-dominated regime. The proof relies on the definitions and constructions developed in Sections A.1–A.2 above.

Theorem 3.1 (Conditional MSE Reduction of Layer 1 Retrieval Correction)**.** *Conditional on $(I_{t,j}, \{h_i^{(t)}\})$, let $\sigma^2 = \text{Var}(w_{t,j}^{std} \mid \cdot)$, $\sigma_{retr}^2 = \text{Var}(\hat{w}_{retr})$, $\Delta = \mathbb{E}[\hat{w}_{retr}] - w_{t,j}^{\dagger}$, and $\rho = \text{Corr}(w_{t,j}^{std}, \hat{w}_{retr})$. Assume the variance-dominated leaf regime in which the conditional Newton variance follows the Lemma 3.1 rate, i.e. $\sigma^2 = \sigma_j^2/(N_{t,j} h_{\min}^2)$. Under Proposition 3.1, there exists a threshold N_{rag}^* (given explicitly by Eq.* (A.4) *below), decreasing in σ_{retr}^2 and Δ^2, such that*

$$\text{MSE}(w_{t,j}^{rag}) < \text{MSE}(w_{t,j}^{std}) \quad \textit{whenever} \quad N_{t,j} < N_{rag}^*.$$

Proof. Framework. Condition throughout on $(I_{t,j}, \{h_i^{(t)}\})$. Under this conditioning, $w_{t,j}^{\text{std}}$ is conditionally unbiased for the conditional Newton step $w_{t,j}^{\dagger}$ (Definition A.1, Section A.6), so $\text{MSE}(w_{t,j}^{\text{std}}) = \sigma^2 := \text{Var}(w_{t,j}^{\text{std}} \mid \cdot)$, equal to the stated $1/N_{t,j}$ variance rate under the theorem's variance-dominated assumption (and bounded above by that rate under the weaker Lemma 3.1 conditions).

Setup. Under the approximate convex combination (Proposition 3.1, Section A.2.4), write $\beta := \tilde{\beta}_{t,j} \in (0, 1)$:

$$w_{t,j}^{\text{rag}} \approx (1 - \beta)\, w_{t,j}^{\text{std}} + \beta\, \hat{w}_{\text{retr}}.$$

MSE of $w_{t,j}^{rag}$. The bias is $\beta\Delta$ and the variance is

$$\text{Var}(w_{t,j}^{\text{rag}}) = (1 - \beta)^2 \sigma^2 + \beta^2 \sigma_{\text{retr}}^2 + 2\beta(1 - \beta)\rho\, \sigma\, \sigma_{\text{retr}},$$

so

$$\text{MSE}(w_{t,j}^{\text{rag}}) = (1 - \beta)^2 \sigma^2 + \beta^2(\sigma_{\text{retr}}^2 + \Delta^2) + 2\beta(1 - \beta)\rho\, \sigma\, \sigma_{\text{retr}}.$$

Reduction condition. We require $\text{MSE}(w_{t,j}^{\text{rag}}) < \sigma^2$:

$$\sigma^2 - \text{MSE}(w_{t,j}^{\text{rag}}) = \beta(2 - \beta)\sigma^2 - \beta^2(\sigma_{\text{retr}}^2 + \Delta^2) - 2\beta(1 - \beta)\rho\, \sigma\, \sigma_{\text{retr}} > 0.$$

Dividing by $\beta > 0$:

$$(2-\beta)\sigma^2 - 2(1-\beta)\rho\,\sigma\,\sigma_{\text{retr}} - \beta(\sigma_{\text{retr}}^2 + \Delta^2) > 0. \tag{A.2}$$

Treating (A.2) as a quadratic inequality in σ with positive leading coefficient $(2-\beta) > 0$, it holds whenever $\sigma > \sigma_*^+$, where σ_*^+ is the larger root:

$$\sigma_*^+ = \frac{(1-\beta)\rho\,\sigma_{\text{retr}} + \sqrt{(1-\beta)^2\rho^2\sigma_{\text{retr}}^2 + (2-\beta)\beta(\sigma_{\text{retr}}^2 + \Delta^2)}}{2-\beta}. \tag{A.3}$$

Sample-count threshold. Under the stated variance-dominated condition, equating $\sigma^2 = \sigma_j^2/(N_{t,j}\,h_{\min}^2)$ to the quadratic threshold $(\sigma_*^+)^2$ gives the exact sample-count threshold for this conditional regime:

$$N_{\text{rag}}^* = \frac{(2-\beta)^2\,\sigma_j^2/h_{\min}^2}{\Big((1-\beta)\rho\,\sigma_{\text{retr}} + \sqrt{(1-\beta)^2\rho^2\sigma_{\text{retr}}^2 + (2-\beta)\beta(\sigma_{\text{retr}}^2 + \Delta^2)}\Big)^2}. \tag{A.4}$$

For all $N_{t,j} < N_{\text{rag}}^*$ under the stated variance-rate condition, $\text{MSE}(w_{t,j}^{\text{rag}}) < \text{MSE}(w_{t,j}^{\text{std}})$. □

Remark A.1 (Consistency of gating with the theoretical threshold)**.** *Inspection of* (A.4) *shows N_{rag}^* is strictly decreasing in σ_{retr}^2 and Δ^2: unreliable retrieval shrinks the regime of guaranteed improvement. This is consistent with the gating mechanism designed in Section A.2.3, which reduces $\alpha_{t,j}$ (and hence β) precisely when these quantities grow.*

A.4 Layer 2: Ancillary Residual Correction and Diagnostic Reweighting

This section presents the complete Layer 2 framework and establishes the asymptotic recovery and safe fallback properties of the full RAGBoost update. The original Layer 2 formulation treats ancillary trees as sources of leaf-local correction signals. To preserve the original logic, this scalar correction remains the canonical analytical form used in Theorem 3.2. We additionally introduce a diagnostic reweighting refinement that keeps the same ancillary trees and gating notation but allows their information to affect the Newton statistics of the main leaf update at the sample level.

A.4.1 Motivation

Layer 1 (Section A.2) addresses Regime I (Variance-dominated) but is structurally unable to resolve Regime II (Bias-dominated): when the memory bank $\mathcal{M}$ contains no relevant entries, the gating mechanism correctly suppresses $\alpha_{t,j}$, and the update falls back to the standard weight. Layer 2 injects leaf-local correction signals derived from ancillary trees—trees fitted on the current negative gradients under diverse inductive biases, whose predictions provide correction signals but never enter the ensemble.

The purpose of Layer 2 is not to create an additional ensemble of predictive trees. Rather, the ancillary trees serve as transient residual probes. They identify whether a main-tree leaf contains unresolved residual structure that a single leaf weight would otherwise average away. In the scalar form, this information enters through an additive correction term. In the diagnostic reweighting refinement, this information enters by assigning higher Newton influence to samples that ancillary trees identify as belonging to systematic residual substructure.

A.4.2 Ancillary Tree Induction

At each boosting iteration t, before computing the main tree's leaf weights, we fit P ancillary trees $\{f_t^{(p)}\}_{p=1}^P$ to the current negative gradients $r_i^{(t)} = -g_i^{(t)}$ under user-specified constraints encoding distinct inductive biases. Typical configurations include shallow trees for low-order main effects, deeper trees for high-order interactions, and subsampled feature splits to decorrelate the induced structures [Ho, 1998]. The number P and the specific constraints are configurable hyperparameters.

Crucially, the ancillary trees do not update the ensemble F_{t-1}, do not contribute to the model output at inference time, and are discarded after their leaf predictions are extracted. A formal analysis of the structural distinction from ensemble methods is provided in Section A.7.

A.4.3 Directional Alignment Score

For leaf node j of the main tree, we assess whether ancillary tree p captures residual structure that the main tree's current weight does not. Let $\mathbf{r}_{t,j} = (r_i^{(t)})_{i \in I_{t,j}} \in \mathbb{R}^{N_{t,j}}$ be the negative-gradient vector restricted to leaf j, and $\hat{\mathbf{r}}_{t,j}^{(p)} = (\hat{r}_i^{(p)})_{i \in I_{t,j}} \in \mathbb{R}^{N_{t,j}}$ the ancillary tree's prediction vector restricted to leaf j. The directional alignment score is

$$A_{t,j}^{(p)} = \frac{\langle \mathbf{r}_{t,j},\ \hat{\mathbf{r}}_{t,j}^{(p)} \rangle}{\|\mathbf{r}_{t,j}\|_2\ \|\hat{\mathbf{r}}_{t,j}^{(p)}\|_2 + \epsilon} \in [-1, 1],$$

where $\epsilon > 0$ prevents division by zero. A high $A_{t,j}^{(p)}$ indicates that ancillary tree p is directionally aligned with the negative gradient in leaf j. This is necessary but not always sufficient for detecting leaf-internal residual structure. Therefore, the diagnostic refinement below optionally supplements alignment with an explanation score that measures how much within-leaf residual variation is explained by the ancillary tree.

A.4.4 Gating and Scalar Correction Term

Only ancillary trees with sufficient alignment are admitted. The valid ancillary set is

$$\mathcal{P}_{t,j} = \left\{ p : A_{t,j}^{(p)} > \tau \right\},$$

where $\tau > 0$ is a threshold hyperparameter. The ancillary correction term is

$$\delta_{t,j}^{\text{anc}} = \sum_{p \in \mathcal{P}_{t,j}} \omega_{t,j}^{(p)}\, \bar{r}_{t,j}^{(p)},$$

where $\bar{r}_{t,j}^{(p)} = N_{t,j}^{-1} \sum_{i \in I_{t,j}} \hat{r}_i^{(p)}$ is the mean ancillary prediction within the leaf, and

$$\omega_{t,j}^{(p)} = \frac{A_{t,j}^{(p)}}{\sum_{p' \in \mathcal{P}_{t,j}} A_{t,j}^{(p')}}$$

are alignment-proportional weights. If $\mathcal{P}_{t,j} = \emptyset$, we set $\delta_{t,j}^{\text{anc}} = 0$.

This scalar form is retained as the analytical representation of Layer 2 because it permits a direct MSE comparison with the Layer 1 estimator in Section A.5. The refinement in Section A.4.5 keeps the same ancillary trees and the same valid-set logic, but replaces or supplements the leaf-average scalar with sample-level diagnostic responsibility scores.

A.4.5 Diagnostic Reweighting Refinement of Layer 2

The scalar correction in Section A.4.4 summarizes ancillary information by the leaf mean $\bar{r}_{t,j}^{(p)}$. This is conservative, but it can dilute useful signals when the main tree leaf contains a smaller residual subregion that is important for loss reduction. To preserve the original framework while making the ancillary signal more actionable, we define a compatible diagnostic reweighting refinement.

For each main-tree leaf j, define the within-leaf residual variance

$$V_{t,j} = \frac{1}{N_{t,j}} \sum_{i \in I_{t,j}} \left(r_i^{(t)} - \bar{r}_{t,j} \right)^2, \qquad \bar{r}_{t,j} = \frac{1}{N_{t,j}} \sum_{i \in I_{t,j}} r_i^{(t)}.$$

For ancillary tree p, define its residual-structure explanation score inside leaf j as

$$E_{t,j}^{(p)} = 1 - \frac{\sum_{i \in I_{t,j}} \left(r_i^{(t)} - \hat{r}_i^{(p)} \right)^2}{\sum_{i \in I_{t,j}} \left(r_i^{(t)} - \bar{r}_{t,j} \right)^2 + \epsilon}. \tag{A.5}$$

A large positive $E_{t,j}^{(p)}$ means that ancillary tree p explains within-leaf residual variation beyond the leaf mean. We therefore refine the valid ancillary set as

$$\mathcal{P}_{t,j} = \left\{ p : A_{t,j}^{(p)} > \tau \ \text{ and } \ E_{t,j}^{(p)} > \tau_E \right\}, \tag{A.6}$$

where τ_E is a diagnostic explanation threshold. When the diagnostic refinement is disabled, (A.6) reduces to the original alignment-only definition by taking $\tau_E = -\infty$.

For each accepted ancillary tree, define a sample-level responsibility score

$$s_i^{(p)} = \frac{|\hat{r}_i^{(p)}|}{N_{t,j}^{-1} \sum_{k \in I_{t,j}} |\hat{r}_k^{(p)}| + \epsilon}, \qquad i \in I_{t,j}. \tag{A.7}$$

The accepted ancillary trees are aggregated with diagnostic-proportional weights

$$\omega_{t,j}^{(p)} = \frac{\max(E_{t,j}^{(p)}, 0)}{\sum_{p' \in \mathcal{P}_{t,j}} \max(E_{t,j}^{(p')}, 0) + \epsilon}, \tag{A.8}$$

yielding the aggregated ancillary responsibility

$$s_i^{\text{anc}} = \sum_{p \in \mathcal{P}_{t,j}} \omega_{t,j}^{(p)} s_i^{(p)}. \tag{A.9}$$

If $\mathcal{P}_{t,j} = \emptyset$, we set $s_i^{\text{anc}} = 1$ for all $i \in I_{t,j}$, so no sample-level reweighting is applied.

The ancillary responsibility score is converted into a bounded Newton reweighting factor

$$a_i^{\text{anc}} = 1 + \lambda_{\text{anc}} \cdot \text{clip}\left(s_i^{\text{anc}} - 1,\ 0,\ s_{\max}\right), \tag{A.10}$$

where $\lambda_{\text{anc}} \geq 0$ controls the strength of diagnostic reweighting and $s_{\max}$ prevents individual samples from dominating the update. The reweighted gradients and Hessians are

$$\tilde{g}_i^{(t)} = a_i^{\text{anc}} g_i^{(t)}, \qquad \tilde{h}_i^{(t)} = a_i^{\text{anc}} h_i^{(t)}, \tag{A.11}$$

and the corresponding leaf aggregates are

$$\tilde{G}_{t,j} = \sum_{i \in I_{t,j}} \tilde{g}_i^{(t)}, \qquad \tilde{H}_{t,j} = \sum_{i \in I_{t,j}} \tilde{h}_i^{(t)}. \tag{A.12}$$

The diagnostic-reweighted Layer 2 update is then

$$\boxed{w_{t,j}^{\text{v2}} = -\frac{\tilde{G}_{t,j} + \alpha_{t,j} G_{t,j}^{\text{prior}}}{\tilde{H}_{t,j} + \alpha_{t,j} H_{t,j}^{\text{prior}} + \lambda}.} \tag{A.13}$$

This update is a minimal modification of the original framework: it keeps the same main tree, the same retrieval prior, and the same ancillary trees, but lets ancillary information alter the Newton statistics instead of only adding a leaf-average residual term. When $\lambda_{\text{anc}} = 0$ or $\mathcal{P}_{t,j} = \emptyset$, we have $a_i^{\text{anc}} = 1$, $\tilde{G}_{t,j} = G_{t,j}$, $\tilde{H}_{t,j} = H_{t,j}$, and (A.13) reduces exactly to the Layer 1 retrieval update $w_{t,j}^{\text{rag}}$. If retrieval is also suppressed ($\alpha_{t,j} = 0$), it reduces to the standard weight $w_{t,j}^{\text{std}}$.

The scalar correction in Section A.4.4 and the diagnostic reweighting update in (A.13) are two compatible implementations of the same ancillary principle. The scalar form treats the ancillary output as a leaf-level correction direction; the reweighted form treats it as a sample-level diagnostic signal that reshapes the Newton update. Unless otherwise stated, the theoretical MSE statement in Theorem 3.2 refers to the scalar form, while Proposition A.1 below records the safe-reduction property of the diagnostic-reweighted form.

A.4.6 Unified Leaf Weight Update

Combining all three sources—standard estimate, gradient-space priors, and ancillary correction—the original scalar Layer 2 update is

$$\boxed{w_{t,j}^{\text{v2}} = w_{t,j}^{\text{rag}} + \mu_{t,j}^{\text{anc}}\, \delta_{t,j}^{\text{anc}},} \tag{A.14}$$

where

$$\mu_{t,j}^{\text{anc}} = \frac{\eta}{1 + \alpha_{t,j}\, H_{t,j}^{\text{prior}}} \cdot \frac{|\mathcal{P}_{t,j}|}{P},$$

with global scale $\eta > 0$. This design encodes two principles:

1. Complementarity with Layer 1: the factor $1/(1+\alpha_{t,j}\,H^{\text{prior}}_{t,j})$ ensures that when retrieval is confident ($\alpha_{t,j}$ large), Layer 2 defers; when retrieval is uninformative ($\alpha_{t,j}\approx 0$), Layer 2 takes over.
2. Ancillary consensus: the factor $|\mathcal{P}_{t,j}|/P$ scales the correction by the fraction of ancillary trees passing the alignment or diagnostic threshold.

In implementations using diagnostic reweighting, (A.13) is used in place of (A.14). Both updates preserve the same safe fallback principle: if ancillary evidence is absent, the ancillary contribution vanishes; if retrieval evidence is absent, the retrieval contribution vanishes.

A.4.7 Safe Gain Gate for Diagnostic Reweighting

Because diagnostic reweighting increases the influence of selected samples, we add a conservative local gain gate. Let the unweighted retrieval gain proxy be

$$\text{Gain}^{\text{rag}}_{t,j}=\frac{(G_{t,j}+\alpha_{t,j}G^{\text{prior}}_{t,j})^2}{H_{t,j}+\alpha_{t,j}H^{\text{prior}}_{t,j}+\lambda},$$

and the diagnostic-reweighted gain proxy be

$$\text{Gain}^{\text{diag}}_{t,j}=\frac{(\tilde{G}_{t,j}+\alpha_{t,j}G^{\text{prior}}_{t,j})^2}{\tilde{H}_{t,j}+\alpha_{t,j}H^{\text{prior}}_{t,j}+\lambda}.$$

The diagnostic-reweighted update is accepted only if

$$\text{Gain}^{\text{diag}}_{t,j}>\text{Gain}^{\text{rag}}_{t,j}+\varepsilon_{\text{gain}}, \tag{A.15}$$

where $\varepsilon_{\text{gain}}\geq 0$ is a small tolerance. Otherwise, the model falls back to the original retrieval update, or to the scalar correction pathway when that pathway is explicitly selected. This gate prevents ancillary trees from changing the main update unless their induced weighting improves the local second-order objective proxy.

A.4.8 Asymptotic Recovery and Safe Fallback (Proposition 3.2)

Proposition 3.2 (Asymptotic Recovery and Safe Fallback)**.** *(i) As $N_{t,j}\to\infty$, $w^{v2}_{t,j}\to w^{std}_{t,j}$ under the scalar correction pathway, and $w^{v2}_{t,j}\to w^{std}_{t,j}$ under the diagnostic-reweighting pathway when the ancillary responsibility vanishes or is suppressed by the diagnostic gate asymptotically.*

(ii) When retrieval is uninformative ($\alpha_{t,j}=0$) and $\mathcal{P}_{t,j}=\emptyset$, $w^{v2}_{t,j}=w^{std}_{t,j}$ exactly.

Proof. Part (i): Asymptotic recovery for Layer 1. From the gating definition (Section A.2.3),

$$\alpha_{t,j}=\frac{c}{N^{\gamma}_{t,j}(\tilde{\zeta}^{\text{retr}}_{t,j}+\tilde{\Delta}^2_{t,j}+\epsilon)}.$$

Since $\gamma>0$ and the denominator is bounded below by $\epsilon>0$, we have $\alpha_{t,j}\to 0$, so $w^{\text{rag}}_{t,j}\to w^{\text{std}}_{t,j}$.

Convergence of $w^{std}_{t,j}$. By the law of large numbers, $G_{t,j}/N_{t,j}\xrightarrow{p}\mathbb{E}[g\mid R_{t,j}]$ and $H_{t,j}/N_{t,j}\xrightarrow{p}\mathbb{E}[h\mid R_{t,j}]$. With $\lambda/N_{t,j}\to 0$, $w^{\text{std}}_{t,j}\xrightarrow{p}w^{*}_{t,j}$ (Definition A.1, Section A.6).

Scalar Layer 2 pathway. Assume each ancillary tree has bounded depth such that, as $N_{t,j}\to\infty$, each of its leaves intersecting $R_{t,j}$ contains $\Theta(N_{t,j})$ samples (this holds for fixed-depth trees under standard splitting rules). Then $\bar{r}^{(p)}_{t,j}$ is a sample average over $\Theta(N_{t,j})$ negative gradients whose conditional mean converges to zero whenever the within-leaf residual structure is exhausted by the main update. Therefore $\bar{r}^{(p)}_{t,j}\xrightarrow{p}0$, and consequently $\delta^{\text{anc}}_{t,j}\xrightarrow{p}0$. Since $\mu^{\text{anc}}_{t,j}\leq\eta$ is bounded, $\mu^{\text{anc}}_{t,j}\delta^{\text{anc}}_{t,j}\xrightarrow{p}0$. Combining with Layer 1 gives $w^{\text{v2}}_{t,j}\to w^{\text{std}}_{t,j}$.

Diagnostic-reweighting pathway. If the accepted ancillary diagnostic set becomes empty asymptotically, or if the gain gate (A.15) rejects diagnostic reweighting asymptotically, then $a^{\text{anc}}_i=1$ for all

$i \in I_{t,j}$ and the diagnostic update reduces to $w_{t,j}^{\text{rag}}$, which converges to $w_{t,j}^{\text{std}}$ as shown above. More generally, if $a_i^{\text{anc}} \xrightarrow{p} 1$ uniformly over $i \in I_{t,j}$, then $\tilde{G}_{t,j} - G_{t,j} = o_p(N_{t,j})$ and $\tilde{H}_{t,j} - H_{t,j} = o_p(N_{t,j})$, so (A.13) has the same probability limit as $w_{t,j}^{\text{rag}}$ and hence as $w_{t,j}^{\text{std}}$.

Part (ii): Exact fallback. If $\alpha_{t,j} = 0$, then $G_{t,j}^{\text{prior}}$ and $H_{t,j}^{\text{prior}}$ receive zero weight in the numerator and denominator of $w_{t,j}^{\text{rag}}$, yielding $w_{t,j}^{\text{rag}} = w_{t,j}^{\text{std}}$. If additionally $\mathcal{P}_{t,j} = \emptyset$, then in the scalar pathway $\delta_{t,j}^{\text{anc}} = 0$ by definition, so $w_{t,j}^{\text{v2}} = w_{t,j}^{\text{std}}$. In the diagnostic pathway, $\mathcal{P}_{t,j} = \emptyset$ implies $s_i^{\text{anc}} = 1$ and $a_i^{\text{anc}} = 1$, so $\tilde{G}_{t,j} = G_{t,j}$ and $\tilde{H}_{t,j} = H_{t,j}$. With $\alpha_{t,j} = 0$, (A.13) also gives $w_{t,j}^{\text{v2}} = w_{t,j}^{\text{std}}$. □

Corollary A.1 (Regime-Conditional Improvement and Safe Fallback)**.** *Under the conditions of Theorems 3.1 and 3.2: Layer 1 reduces MSE relative to the standard weight when $N_{t,j} < N_{rag}^*$ (Theorem 3.1, Section A.3); Layer 2 further reduces MSE relative to the Layer 1 weight when Condition* (A.16) *holds (Theorem 3.2, Section A.5). When diagnostic reweighting is used, it is applied only when the diagnostic gain gate* (A.15) *is satisfied. When neither correction is reliable, both gating mechanisms suppress their respective correction terms and the update reduces to $w_{t,j}^{v2} = w_{t,j}^{std}$—the standard GBDT leaf weight.*

Proposition A.1 (Local Descent Proxy for Diagnostic Reweighting)**.** *Fix a leaf j and condition on the current tree structure, retrieval quantities, and ancillary diagnostic weights. Let $\text{Gain}_{t,j}^{rag}$ and $\text{Gain}_{t,j}^{diag}$ be defined as in Section A.4.7. If* (A.15) *holds, then the diagnostic-reweighted update yields a strictly larger local second-order gain proxy than the unweighted retrieval update by at least ε_{gain}.*

Proof. The statement follows directly from the acceptance criterion (A.15). The purpose of the proposition is not to claim global loss dominance, but to formalize the safeguard used by the diagnostic pathway: the ancillary-induced weighting is accepted only when it improves the local Newton gain proxy relative to the Layer 1 update. □

Remark A.2 (Interpretation of diagnostic reweighting)**.** *The diagnostic reweighting refinement changes the training dynamics rather than the deployed model class. Ancillary trees still do not appear in the final prediction function. Their role is to detect leaf-internal residual structure and increase the Newton influence of samples that are likely to represent systematic errors rather than irreducible noise. This makes Layer 2 more directly responsive to Regime II (Bias-dominated) while preserving the original safe fallback behavior.*

A.5 Proof of Theorem 3.2 (Layer 2 Conditional MSE Improvement)

We now establish that Layer 2 further reduces MSE when the ancillary correction is appropriately aligned with the Layer 1 estimation error. The notation and constructions used here are developed in Sections A.1–A.4 above. The theorem is stated for the scalar correction pathway, which provides a clean conditional MSE comparison. The diagnostic reweighting pathway in Section A.4.5 is governed by the local gain gate in Proposition A.1 and can be viewed as an implementation refinement of how ancillary information enters the Newton update.

Theorem 3.2 (Layer 2 Conditional MSE Improvement)**.** *Conditional on $(I_{t,j}, \{h_i^{(t)}\})$ and the gating quantities $(\alpha_{t,j}, \mu_{t,j}^{anc}, \mathcal{P}_{t,j})$, let $e_{t,j}^{rag} = w_{t,j}^{rag} - w_{t,j}^{\dagger}$ with $\sigma_{rag}^2 = \text{Var}(e_{t,j}^{rag})$, $\sigma_{anc}^2 = \text{Var}(\delta_{t,j}^{anc})$, and $\Delta_{anc} = \mathbb{E}[\delta_{t,j}^{anc}] + \mathbb{E}[e_{t,j}^{rag}]$. Then*

$$\text{MSE}(w_{t,j}^{v2}) < \text{MSE}(w_{t,j}^{rag})$$

holds whenever

$$\text{Corr}\big(e_{t,j}^{rag},\ \delta_{t,j}^{anc}\big) < -\frac{\mu_{t,j}^{anc}\big(\sigma_{anc}^2 + \Delta_{anc}^2\big) + 2\,\mathbb{E}[e_{t,j}^{rag}]\,\mathbb{E}[\delta_{t,j}^{anc}]}{2\,\sigma_{rag}\,\sigma_{anc}}. \tag{A.16}$$

Proof. Step 1 (Decomposition). Under the stated conditioning, $\mu_{t,j}^{\text{anc}}$ is deterministic. With $e_{t,j}^{\text{rag}} = w_{t,j}^{\text{rag}} - w_{t,j}^{\dagger}$:

$$w_{t,j}^{\text{v2}} - w_{t,j}^{\dagger} = e_{t,j}^{\text{rag}} + \mu_{t,j}^{\text{anc}}\,\delta_{t,j}^{\text{anc}},$$

so

$$\text{MSE}(w_{t,j}^{\text{v2}}) = \mathbb{E}\Big[\big(e_{t,j}^{\text{rag}} + \mu_{t,j}^{\text{anc}}\,\delta_{t,j}^{\text{anc}}\big)^2\Big] = \text{MSE}(w_{t,j}^{\text{rag}}) + 2\mu_{t,j}^{\text{anc}}\,\mathbb{E}[e_{t,j}^{\text{rag}}\,\delta_{t,j}^{\text{anc}}] + (\mu_{t,j}^{\text{anc}})^2\,\mathbb{E}[(\delta_{t,j}^{\text{anc}})^2].$$

Step 2 (Reduction condition). MSE improves iff

$$2\,\mathbb{E}[e_{t,j}^{\text{rag}}\,\delta_{t,j}^{\text{anc}}] + \mu_{t,j}^{\text{anc}}\,\mathbb{E}[(\delta_{t,j}^{\text{anc}})^2] < 0. \tag{A.17}$$

Step 3 (Covariance decomposition). Let $b_{\text{rag}} = \mathbb{E}[e_{t,j}^{\text{rag}}]$ denote the Layer 1 bias. Then

$$\mathbb{E}[e_{t,j}^{\text{rag}}\,\delta_{t,j}^{\text{anc}}] = \text{Cov}(e_{t,j}^{\text{rag}},\,\delta_{t,j}^{\text{anc}}) + b_{\text{rag}}\,\mathbb{E}[\delta_{t,j}^{\text{anc}}],$$

and $\mathbb{E}[(\delta_{t,j}^{\text{anc}})^2] = \sigma_{\text{anc}}^2 + (\mathbb{E}[\delta_{t,j}^{\text{anc}}])^2$. Substituting into (A.17):

$$2\,\sigma_{\text{rag}}\,\sigma_{\text{anc}}\,\text{Corr}(e_{t,j}^{\text{rag}},\,\delta_{t,j}^{\text{anc}}) + 2\,b_{\text{rag}}\,\mathbb{E}[\delta_{t,j}^{\text{anc}}] + \mu_{t,j}^{\text{anc}}\big(\sigma_{\text{anc}}^2 + (\mathbb{E}[\delta_{t,j}^{\text{anc}}])^2\big) < 0.$$

Rearranging:

$$\text{Corr}(e_{t,j}^{\text{rag}},\,\delta_{t,j}^{\text{anc}}) < -\frac{\mu_{t,j}^{\text{anc}}\big(\sigma_{\text{anc}}^2 + (\mathbb{E}[\delta_{t,j}^{\text{anc}}])^2\big) + 2\,b_{\text{rag}}\,\mathbb{E}[\delta_{t,j}^{\text{anc}}]}{2\,\sigma_{\text{rag}}\,\sigma_{\text{anc}}}.$$

With $\Delta_{\text{anc}} = \mathbb{E}[\delta_{t,j}^{\text{anc}}] + b_{\text{rag}}$: in the regime where Layer 1 is approximately unbiased ($b_{\text{rag}} \approx 0$, which holds when retrieval is accurate or fully suppressed by the gate), we have $\mathbb{E}[\delta_{t,j}^{\text{anc}}] \approx \Delta_{\text{anc}}$ and the condition simplifies to (A.16). □

Remark A.3 (Design motivation for the alignment and diagnostic filters)**.** *Condition* (A.16) *requires the ancillary correction $\delta_{t,j}^{anc}$ to be negatively correlated with the Layer 1 estimation error $e_{t,j}^{rag}$. The directional alignment filter (Sections A.4.3–A.4.4) is designed as a practical proxy for this condition: ancillary trees are admitted only when their predictions align with the current negative gradient $r^{(t)} = -g^{(t)}$, which is the descent direction of the loss evaluated at the current model. The diagnostic explanation score (Section A.4.5) strengthens this proxy by requiring ancillary trees to explain residual variation within the main-tree leaf rather than merely having a positive average direction. Under the heuristic that high alignment plus positive residual-structure explanation indicates systematic residual structure rather than noise, the ancillary pathway tends to move the update toward unresolved error modes. We emphasize that this is a design rationale, not a formal implication of Theorem 3.2; the theorem provides the abstract MSE condition, while the alignment and diagnostic filters are empirically motivated mechanisms for approximately satisfying it.*

A.6 Population-Optimal Leaf Weight

Definition A.1 (Population-optimal leaf weight and conditional Newton step)**.** *For leaf region $R_{t,j}$, the population-optimal leaf weight is*

$$w_{t,j}^{*} = -\frac{\mathbb{E}[g_i^{(t)} \mid x_i \in R_{t,j}]}{\mathbb{E}[h_i^{(t)} \mid x_i \in R_{t,j}]},$$

with expectation over the data-generating distribution conditional on the leaf assignment. This is the probability limit of $w_{t,j}^{std}$ as $N_{t,j} \to \infty$ (with $\lambda/N_{t,j} \to 0$), corresponding to the optimal Newton step for the expected loss within the leaf region.

The conditional Newton step, conditioned on $I_{t,j}$ and $\{h_i^{(t)}\}_{i \in I_{t,j}}$, is

$$w_{t,j}^{\dagger} = -\frac{\mathbb{E}[G_{t,j} \mid I_{t,j}, \{h_i^{(t)}\}]}{H_{t,j} + \lambda}.$$

Under this conditioning, $w_{t,j}^{\text{std}}$ is conditionally unbiased for $w_{t,j}^{\dagger}$, so $\text{MSE}(w_{t,j}^{\text{std}}) = \text{Var}(w_{t,j}^{\text{std}} \mid \cdot)$ in the conditional analysis of Theorems 3.1–3.2. As $\lambda/N_{t,j} \to 0$, $w_{t,j}^{\dagger} \to w_{t,j}^{*}$, and the two estimation targets coincide.

The distinction between $w_{t,j}^{\dagger}$ and $w_{t,j}^{*}$ matters for finite λ: $w_{t,j}^{\text{std}}$ is exactly unbiased for $w_{t,j}^{\dagger}$ but not for $w_{t,j}^{*}$ (there is an $\mathcal{O}(\lambda/N_{t,j})$ regularization bias). All conditional MSE statements in Theorems 3.1–3.2 are made relative to $w_{t,j}^{\dagger}$; the connection to $w_{t,j}^{*}$ holds in the low-regularization regime.

For the diagnostic reweighting pathway, (A.13) can be viewed as optimizing a diagnostically weighted local Newton objective. Its target is therefore a reweighted conditional Newton step during the current iteration. The safe fallback and asymptotic arguments in Proposition 3.2 ensure that when diagnostic evidence vanishes or is suppressed, this target collapses back to the original conditional Newton step and hence to the population-optimal leaf weight in the same low-regularization asymptotic regime.

A.7 Distinction from Ensemble Methods

The ancillary mechanism of RAGBoost may superficially resemble model ensembling, since gradient boosting is itself an additive ensemble. The key structural difference is that in an ensemble, each base model is a self-contained predictor retained at inference time and contributes globally to every prediction. In RAGBoost, by contrast, ancillary trees are ephemeral diagnostic probes: they are fitted at each iteration to the current residual, used to compute correction signals, and then discarded. The final prediction function remains a standard additive tree ensemble $F(x) = \sum_t f_t(x)$, containing no trace of the ancillary trees.

Regarding self-contained predictive components. Ancillary trees are not self-contained predictors. Their outputs do not enter $F(x)$ directly; instead, they contribute to leaf-local corrections (in the scalar pathway) or sample-level diagnostic weights (in the diagnostic reweighting pathway). In neither case does an ancillary tree produce a standalone model-level prediction.

Regarding persistence and global scope. Ancillary trees are constructed, used, and discarded within a single boosting iteration. They impose no inference-time memory or computational cost. Furthermore, their influence is conditional rather than global: corrections are applied only to leaves where ancillary trees pass alignment and diagnostic filters, and where the local gain gate accepts the reweighted update. The scope of combination is restricted to a subset of leaves satisfying data-dependent admissibility criteria.

Regarding independence of construction. Ancillary trees are fitted to $r^{(t)} = -g^{(t)}$, which depends on the entire boosting history F_{t-1}. They are not independently constructed predictive models; their role is to probe the residual structure of the current boosting state under diverse inductive biases. The appropriate analogy is therefore control variates in Monte Carlo estimation [Nelson, 1990]: auxiliary computations that provide side information during estimation but never appear in the estimator's final functional form. RAGBoost uses P disposable diagnostic probes to refine leaf weights within each boosting iteration, leaving the model's inference-time structure and cost identical to that of standard gradient boosting.

B Interpretability Framework for RAGBoost

The RAGBoost framework introduces structured diagnostic signals that extend beyond standard GBDT interpretability. The two-layer correction mechanism—Layer 1 retrieval-augmented empirical shrinkage and Layer 2 ancillary residual correction—produces a layered interpretability decomposition richer than feature importance alone. Because the retrieval mechanism explicitly tracks which historical leaves inform each prediction, and the ancillary trees diagnose whether leaf-internal residual structure remains unresolved, the framework provides actionable transparency at multiple granularities.

B.1 Two-Layer Correction Diagnostics: Leaf State Classification

Every leaf node (t, j) in the training process can be classified into one of several correction states based on the Layer 1 and Layer 2 diagnostics. This classification applies to leaves; every sample i inherits the state of the leaf $(t, j(i))$ containing it at iteration t.

State S — Standard Update. The leaf bypasses both correction layers: the admission criterion (Section A.2.6) rejects retrieval activation, and no ancillary trees pass the alignment threshold. The leaf receives the standard GBDT weight $w_{t,j}^{\text{std}}$ exactly. This indicates a well-behaved leaf with sufficient sample count and low residual heterogeneity.

State R — Retrieval-Augmented Shrinkage (Layer 1 Only). The leaf is admitted to retrieval ($\text{admit}(j) = 1$) and receives a non-trivial retrieval-based, node-conditional prior, but Layer 2 ancillary correction is inactive ($\mathcal{P}_{t,j} = \emptyset$ or diagnostic reweighting is disabled). The leaf weight is $w_{t,j}^{\text{rag}}$ (Section A.2.3). This state identifies variance-dominated leaves where historical borrowing stabilizes the estimate.

State A — Scalar Ancillary Correction (Layer 1 + Layer 2 Scalar). The leaf receives both retrieval augmentation and the scalar ancillary correction term $\delta^{\text{anc}}_{t,j}$. The final weight is $w^{\text{v2}}_{t,j} = w^{\text{rag}}_{t,j} + \mu^{\text{anc}}_{t,j}\,\delta^{\text{anc}}_{t,j}$ (Eq. (A.14)). This state signals that ancillary trees detected unresolved residual structure within the leaf that the leaf-mean correction can address.

State D — Diagnostic Reweighting (Layer 1 + Layer 2 Diagnostic). The leaf passes the diagnostic gain gate (Section A.4.7) and receives the diagnostic reweighting update $w^{\text{v2}}_{t,j}$ (Eq. (A.13)) with reweighted Newton statistics $(\tilde{G}_{t,j}, \tilde{H}_{t,j})$. This is the strongest correction state: ancillary trees identified systematic residual substructure that warranted sample-level reweighting.

State F — Fallback. Retrieval is admitted but the gating mechanism suppresses $\alpha_{t,j}$ to near-zero due to high retrieval uncertainty, and ancillary trees do not provide valid correction. The leaf effectively falls back to $w^{\text{std}}_{t,j}$, consistent with the safe fallback property (Proposition 3.2). This state flags isolated or inconsistent regions of gradient space.

Formal Definition. For leaf (t, j), define:

$$\text{State}(t,j) = \begin{cases} S & \text{if } \text{admit}(j) = 0 \text{ and } \mathcal{P}_{t,j} = \emptyset \\ R & \text{if } \text{admit}(j) = 1,\ \alpha_{t,j} > 0,\ \mathcal{P}_{t,j} = \emptyset \\ A & \text{if } \text{admit}(j) = 1,\ \mathcal{P}_{t,j} \neq \emptyset, \text{ scalar pathway active} \\ D & \text{if } \text{admit}(j) = 1,\ \mathcal{P}_{t,j} \neq \emptyset, \text{ diagnostic gate (A.15) passed} \\ F & \text{if } \text{admit}(j) = 1,\ \alpha_{t,j} \approx 0,\ \mathcal{P}_{t,j} = \emptyset \end{cases}$$

The interpretability output is a state label for every leaf at every iteration, and a per-sample mode stability score measuring the fraction of iterations each sample spends in non-standard states (R, A, D). Samples with high stability in State D represent systematic residual substructure that the main tree's splits fail to isolate.

B.2 Retrieval-Based Consistency Score

For each sample i, the retrieval process produces a consistency score that quantifies how typical its gradient structure is relative to historical observations.

Definition. The consistency score for sample i at iteration t is:

$$\text{CS}_t(i) = \frac{1}{1 + \tilde{\Delta}^2_{t,j(i)}} \cdot \left(1 - \frac{\tilde{\zeta}^{\text{retr}}_{t,j(i)}}{1+\epsilon}\right) \cdot \mathbb{1}[\text{admit}(j(i)) = 1],$$

where $\tilde{\Delta}^2$ is the normalized geometric drift and $\tilde{\zeta}^{\text{retr}}$ is the normalized retrieved-neighbor weight variance (Section A.2.5). The indicator ensures the score is defined only when retrieval is activated. The score is high when retrieved neighbors are close (low $\tilde{\Delta}^2$) and consistent (low $\tilde{\zeta}^{\text{retr}}$), and zero when the leaf bypasses retrieval entirely.

The aggregated score $\overline{\text{CS}}(i) = \frac{1}{T}\sum_t \text{CS}_t(i)$ produces a per-sample typicality measure. Low-CS samples among those admitted to retrieval are candidates for domain review: they may represent data errors, distributional shift, or genuine rare events. Samples with CS consistently at zero are in well-behaved leaves where retrieval is unnecessary—a positive signal of local stability.

B.3 Decomposed Feature Importance

Standard GBDT feature importance computes split-frequency or gain-based importance aggregated over all trees and all samples. RAGBoost enables a finer decomposition.

State-Conditional Feature Importance. Since every leaf is labeled with a correction state (Section B.1), feature importance can be computed conditioned on state:

$$\phi^{(S)}_f = \sum_t \sum_{j:\text{State}(t,j)=S} \Delta\mathcal{L}_{t,j,f}, \qquad \phi^{(R)}_f = \sum_t \sum_{j:\text{State}(t,j)=R} \Delta\mathcal{L}_{t,j,f},$$

$$\phi_f^{(A)} = \sum_t \sum_{j:\text{State}(t,j)=A} \Delta\mathcal{L}_{t,j,f}, \qquad \phi_f^{(D)} = \sum_t \sum_{j:\text{State}(t,j)=D} \Delta\mathcal{L}_{t,j,f},$$

where $\Delta\mathcal{L}_{t,j,f}$ is the gain from splits on feature f in tree t along paths leading to leaves with the specified state. This yields four feature importance vectors $\phi^{(S)}, \phi^{(R)}, \phi^{(A)}, \phi^{(D)} \in \mathbb{R}^p$, revealing which features drive predictions for well-behaved leaves vs. retrieval-stabilized leaves vs. ancillary-corrected leaves vs. diagnostic-reweighted leaves. Features with high $\phi^{(D)}$ but low $\phi^{(S)}$ are "residual-structure drivers"—features that govern systematic substructure the main tree fails to isolate. This decomposition is not available from any standard GBDT.

Retrieval-Contribution Decomposition. For a given leaf, the Layer 1 retrieval-augmented prediction decomposes into contributions from local evidence vs. retrieved prior (Proposition 3.1):

$$w_{t,j}^{\text{rag}} = a_{t,j} \cdot w_{t,j}^{\text{std}} + b_{t,j} \cdot \hat{w}_{\text{retr}}.$$

The retrieval reliance ratio for sample i across all iterations is:

$$\text{RR}(i) = \frac{\sum_t |b_{t,j(i)} \hat{w}_{\text{retr}}|}{\sum_t |w_{t,j(i)}^{\text{v2}}|}.$$

High $\text{RR}(i)$ indicates that the prediction for sample i relies heavily on historical precedent rather than local gradient evidence—a flag for reduced local confidence. When RR is near zero, the sample's predictions are driven primarily by its own local data.

B.4 Ancillary Diagnostics: Residual Structure Discovery

Layer 2 ancillary trees provide interpretability signals that are unique to RAGBoost and unavailable from any standard GBDT.

Ancillary Explanation Scores. For each leaf (t, j) and each accepted ancillary tree $p \in \mathcal{P}_{t,j}$, the residual-structure explanation score (Section A.4.5, Eq. (A.5))

$$E_{t,j}^{(p)} = 1 - \frac{\sum_{i \in I_{t,j}} (r_i^{(t)} - \hat{r}_i^{(p)})^2}{\sum_{i \in I_{t,j}} (r_i^{(t)} - \bar{r}_{t,j})^2 + \epsilon}$$

measures how much within-leaf residual variation the ancillary tree explains beyond the leaf mean. High $E_{t,j}^{(p)}$ indicates that ancillary tree p captures systematic residual substructure inside the main-tree leaf.

Sample Responsibility and Diagnostic Influence. When diagnostic reweighting is active, each sample i receives an aggregated ancillary responsibility score s_i^{anc} (Section A.4.5, Eq. (A.9)) and a Newton reweighting factor a_i^{anc} (Eq. (A.10)). Samples with $s_i^{\text{anc}} \gg 1$ are identified by ancillary trees as belonging to systematic residual substructure; samples with $s_i^{\text{anc}} \approx 1$ are consistent with the leaf-mean pattern.

Ancillary Alignment Analysis. The directional alignment score $A_{t,j}^{(p)}$ (Section A.4.3) measures cosine similarity between the negative gradient and ancillary tree p's predictions within leaf j. Systematic patterns—e.g., a particular ancillary configuration consistently aligning with high-residual leaves—indicate persistent structural mismatches between the main tree's inductive bias and the data.

B.5 SHAP Compatibility and Enhanced Attribution

Direct SHAP Compatibility. The RAGBoost update produces a scalar leaf weight $w_{t,j}^{\text{v2}}$ for each leaf in each tree. Since the tree structure is unchanged (RAGBoost only modifies leaf weights, not splits), SHAP's TreeExplainer algorithm applies directly: it traverses the tree structure and distributes the leaf weight across features according to their contribution to routing sample i to its leaf. SHAP values are computed identically to standard GBDTs, with the only difference that the leaf values are $w_{t,j}^{\text{v2}}$ instead of $w_{t,j}^{\text{std}}$.

Source-Decomposed SHAP Values (Scalar Pathway). Under the scalar correction pathway, $w_{t,j}^{\text{v2}} = w_{t,j}^{\text{rag}} + \mu_{t,j}^{\text{anc}}\,\delta_{t,j}^{\text{anc}}$, and since $w_{t,j}^{\text{rag}} = a_{t,j}\,w_{t,j}^{\text{std}} + b_{t,j}\,\hat{w}_{\text{retr}}$, SHAP values can be linearly decomposed:

$$\phi_f^{\text{SHAP}}(i) = a_{t,j} \cdot \phi_f^{\text{SHAP,local}}(i) + b_{t,j} \cdot \phi_f^{\text{SHAP,retr}}(i) + \mu_{t,j}^{\text{anc}} \cdot \phi_f^{\text{SHAP,anc}}(i),$$

where each component represents the feature's contribution through a different evidence source. This decomposition allows practitioners to answer: "Is this feature important because of the data I have for this sample, because of historical precedent from similar leaves, or because ancillary trees detected residual structure?"

Diagnostic-Pathway SHAP Notes. Under the diagnostic reweighting pathway, the leaf weight $w_{t,j}^{\text{v2}}$ (Eq. (A.13)) is computed from reweighted Newton statistics $(\tilde{G}_{t,j}, \tilde{H}_{t,j})$. The SHAP decomposition still applies to the final scalar leaf weight, but the "local" component now implicitly incorporates the diagnostic reweighting. For full transparency, the reweighting factor a_i^{anc} can be reported alongside SHAP values.

B.6 Retrieval Provenance: Tracing the Evidence Chain

For each prediction, RAGBoost can provide a retrieval provenance trace: the specific historical leaf nodes that informed the prediction, along with their similarity scores and original contexts. The per-leaf retrieval record is:

$$\text{Provenance}(t,j) = \left\{ (k, \pi_{jk}, t^{(k)}, j^{(k)}, w^{(k)}, \rho_{jk}) : k = 1, \ldots, K \right\},$$

where for each retrieved neighbor k: π_{jk} is the attention weight, $(t^{(k)}, j^{(k)})$ identifies the source tree and leaf, $w^{(k)}$ is the historical leaf weight, and ρ_{jk} is the Euclidean distance in state space. For a given sample, a practitioner can trace which historical training configurations influenced the current prediction, providing an evidence-based justification akin to case-based reasoning. In high-risk domains, this provenance chain serves as an audit trail.

B.7 Case Study

To illustrate how the proposed interpretability signals can be used in practice, we provide a case study on DAG-NoisyRare (with detailed data descriptions provided in Appendix E).

Figure B.1 reports the global SHAP summary plot computed from the final scalar leaf values of RAGBoost. Since RAGBoost does not modify the learned tree structure and only changes leaf values, standard TreeSHAP-style attribution remains applicable. The global SHAP distribution identifies the dominant features controlling the overall prediction function, providing a conventional model-level explanation comparable to standard GBDT interpretation.

Figure B.2 further decomposes feature attribution by correction state. Each sample inherits the correction state of the leaves it visits across boosting iterations, and samples can therefore be grouped according to whether their predictions are mostly governed by standard updates or by non-standard correction states. Compared with ordinary global SHAP, this state-conditional view reveals which features are associated with locally stable regions and which features become important when retrieval-augmented empirical shrinkage or ancillary residual correction is activated. This distinction is specific to RAGBoost: standard GBDTs can report feature importance, but they do not explain whether a feature is important in a stable leaf, a retrieval-corrected leaf, or an ancillary-corrected leaf.

Figure B.3 shows a local SHAP waterfall plot for a representative high-diagnostic sample selected by retrieval reliance and non-standard state stability. The plot explains the final prediction for this sample using the same SHAP interface as standard tree ensembles, while the accompanying RAGBoost diagnostics indicate why the sample is not merely a typical prediction case. In particular, high retrieval reliance suggests that the prediction depends substantially on historical leaf evidence, and high non-standard state stability indicates that the sample repeatedly enters leaves requiring correction.

Figure B.4 summarizes the RAGBoost-specific diagnostic space by plotting retrieval reliance against the retrieval consistency score, with samples colored by non-standard state stability. This view separates locally stable samples from samples that require repeated correction or exhibit weak retrieval

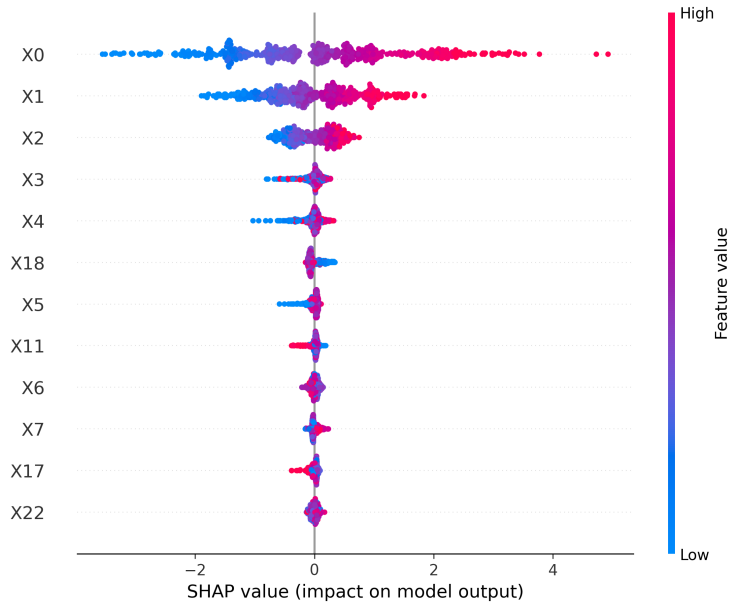


Figure B.1: Global SHAP summary plot on DAG-NoisyRare. Since RAGBoost preserves the tree structure and modifies only scalar leaf values, standard SHAP-style attribution remains applicable to the final model.

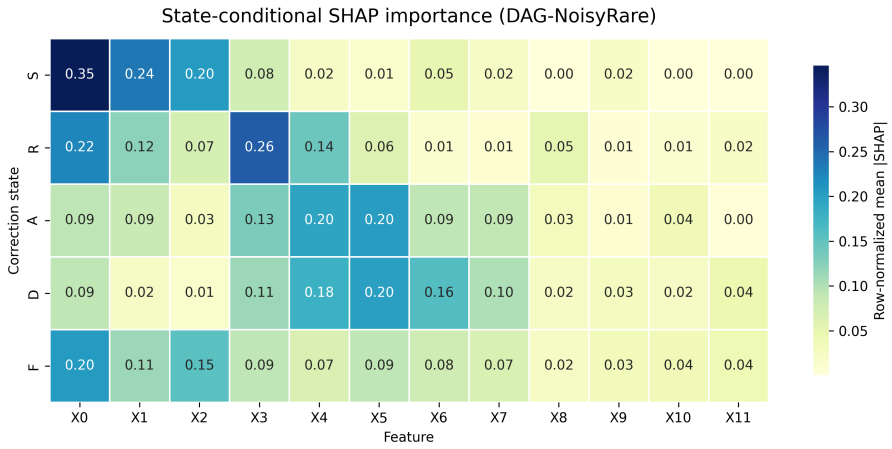


Figure B.2: State-conditional SHAP importance on DAG-NoisyRare. Mean absolute SHAP values are grouped by the dominant correction state of each sample.

consistency. As a result, the interpretation is not limited to feature attribution alone: RAGBoost also provides an evidence-level explanation of when and why a prediction relies on historical leaf states or ancillary residual correction. Together, these results demonstrate that RAGBoost is compatible with standard SHAP analysis while offering additional state-aware and evidence-aware interpretability signals.

B.8 Summary

The following interpretability signals are structurally available from the RAGBoost framework:

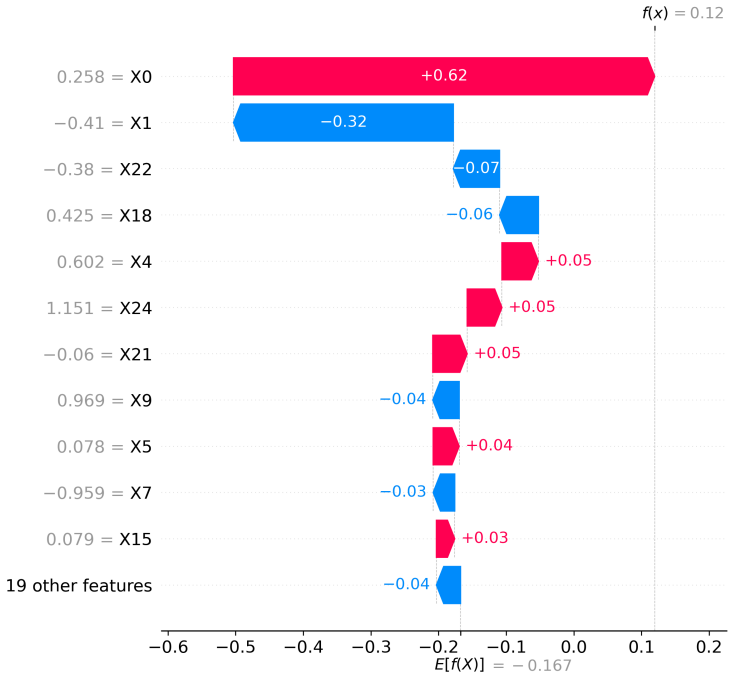


Figure B.3: Local SHAP waterfall plot for a high-diagnostic sample on DAG-NoisyRare.

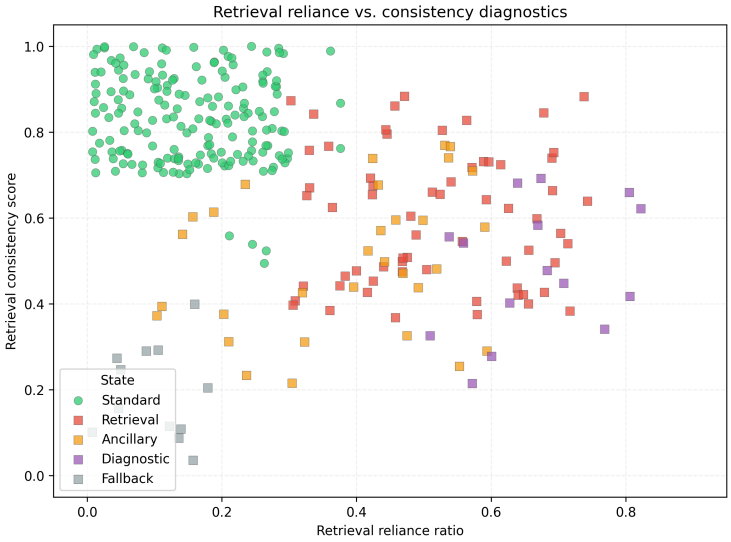


Figure B.4: Retrieval reliance and consistency diagnostics on DAG-NoisyRare.

Signal	Description	Use Case
Leaf state classification	Correction state (S/R/A/D/F) per leaf per iteration	Model debugging, training dynamics
Sample state stability	Fraction of iterations in non-standard states per sample	Identify systematically hard samples
Consistency scores	Per-sample CS values	Anomaly/noise detection
State-conditional feature importance	$\phi^{(S)}, \phi^{(R)}, \phi^{(A)}, \phi^{(D)}$ vectors	State-specific feature analysis
Retrieval reliance ratios	Per-sample RR values	Confidence assessment
Ancillary explanation scores	$E_{t,j}^{(p)}$ per leaf per ancillary tree	Residual structure type discovery
Sample diagnostic influence	$(s_i^{\text{anc}}, a_i^{\text{anc}})$ per sample	Identify influential outliers
Ancillary alignment patterns	$A_{t,j}^{(p)}$ across leaves and trees	Persistent bias detection
Source-decomposed SHAP	SHAP values partitioned by evidence source	Evidence-source attribution
Retrieval provenance	Per-leaf retrieval traces	Audit trail, case-based reasoning
Gating dynamics	$\alpha_{t,j}, \mu_{t,j}^{\text{anc}}$, admit reason per leaf	Retrieval and correction activation analysis

Compared to standard GBDTs. Standard feature importance is a single vector aggregated over all data. RAGBoost provides state-conditional importance, retrieval provenance, ancillary residual-structure discovery, and automatic hard-sample identification—none of which are available from XGBoost, LightGBM, or CatBoost.

Compared to deep tabular models. Neural tabular models are largely opaque: their interpretability relies on post-hoc methods applied to a black-box function. RAGBoost's interpretability is structural: the retrieval mechanism explicitly records which evidence informed each prediction, the gating values quantify uncertainty, and the ancillary trees diagnose what residual structure the main model missed. This is particularly valuable in regulated domains where audit requirements demand traceable evidence chains.

Compared to representation-level retrieval methods. TabR [Gorishniy et al., 2023] retrieves similar training instances in learned embedding space for prediction-level fusion; the embedding space itself is opaque. RAGBoost's retrieval operates in an interpretable gradient-statistics space: the state vectors have known semantic meaning (feature statistics, gradient magnitudes, tree depth), and the retrieval provenance directly identifies which historical training configurations contributed to each update. Moreover, TabR has no analogue to Layer 2 ancillary diagnostics.

C Computational Complexity and Runtime Analysis

This appendix analyzes the computational overhead of RAGBoost from both theoretical and empirical perspectives. The goal is not to enumerate every low-level operation, but to identify the dominant sources of additional cost introduced by the two correction mechanisms described in Appendix A: retrieval-augmented empirical shrinkage and ancillary residual correction.

C.1 Theoretical Complexity

Let n denote the number of training samples, d the feature dimension, T the number of boosting iterations, L the number of leaves in each main tree, q the dimension of the leaf state vector, K the number of retrieved neighbors, $M_t = |\mathcal{M}_t|$ the memory-bank size before iteration t, and P the number of ancillary trees. Let $S_t \leq L$ denote the number of leaves admitted to the retrieval pipeline at iteration t by the criterion in Eq. (A.1). We denote the cost of fitting one standard boosting tree by $\mathcal{C}_{\mathrm{tree}}(n, d)$, and the cost of fitting one ancillary tree by $\mathcal{C}_{\mathrm{anc}}(n, d)$.

For a standard second-order GBDT model, each boosting iteration requires gradient computation, tree construction, and leaf-wise Newton aggregation. Its dominant per-iteration cost can be written as:

$$\mathcal{O}\left(\mathcal{C}_{\mathrm{tree}}(n, d) + n\right).$$

RAGBoost preserves this main-tree training procedure and introduces two additional sources of training-time cost.

Retrieval-Augmented Shrinkage Layer. For each admitted leaf, RAGBoost constructs a state vector $s_{t,j}$ and retrieves K similar historical leaf entries from the memory bank $\mathcal{M}_t$. With approximate nearest-neighbor (ANN) retrieval, the dominant retrieval cost at iteration t is denoted by $S_t T_{\mathrm{ANN}}(M_t, q, K)$:

$$\mathcal{O}(S_t T_{\mathrm{ANN}}(M_t, q, K)).$$

The lower-order cost of aggregating the top-K neighbors is included in $T_{\mathrm{ANN}}(M_t, q, K)$. Thus, Layer 1 mainly introduces an ANN memory-query overhead. This overhead is controlled by two design choices: only admitted leaves activate retrieval, so S_t can be much smaller than L; and the memory bank can be capped at $M_{\max}$, which bounds the first argument of T_{ANN}.

Ancillary Correction Layer. At each boosting iteration, RAGBoost fits P ancillary trees on the current pseudo-residuals. These trees are used only as training-time diagnostic probes and are not retained for inference. The dominant ancillary cost is:

$$\mathcal{O}\left(P\mathcal{C}_{\mathrm{anc}}(n, d)\right).$$

After the ancillary trees are fitted, computing alignment scores, diagnostic explanation scores, and sample-level responsibilities requires linear accumulation over samples, with cost:

$$\mathcal{O}(Pn).$$

Therefore, the main additional cost of Layer 2 is the fitting of P transient ancillary trees, plus a linear sample-level diagnostic pass. If diagnostic reweighting is enabled, computing a_i^{anc}, $\tilde{G}_{t,j}$, and $\tilde{H}_{t,j}$ requires only another linear pass and does not change the asymptotic order.

Combining the standard boosting cost with the two RAGBoost-specific components, the total training complexity can be summarized as:

$$\boxed{\mathcal{O}\left(T\mathcal{C}_{\mathrm{tree}}(n,d) + TP\mathcal{C}_{\mathrm{anc}}(n,d) + TPn + \sum_{t=1}^{T} S_t T_{\mathrm{ANN}}(M_t, q, K)\right).}$$

This expression shows that the dominant overheads are localized in two terms: the retrieval term $\sum_{t=1}^{T} S_t T_{\mathrm{ANN}}(M_t, q, K)$, introduced by Layer 1, and the ancillary-tree term $TP\mathcal{C}_{\mathrm{anc}}(n,d)$, introduced by Layer 2. Other operations, such as computing gates, aggregating retrieved priors, evaluating gain proxies, and updating leaf weights, are lower-order relative to tree construction, memory retrieval, and ancillary-tree fitting.

At inference time, RAGBoost has the same complexity as standard GBDT. The deployed model remains:

$$F(x) = \sum_{t=1}^{T} f_t(x),$$

where f_t is the main boosting tree. The memory bank and all ancillary trees are discarded after training. If each tree has depth at most D, the inference cost for n_{test} test samples is:

$$\boxed{\mathcal{O}(n_{\mathrm{test}} T D),}$$

which is identical to a standard additive tree ensemble with the same number and depth of trees.

C.2 Empirical Runtime Results

To complement the theoretical analysis, we report the measured training runtime of RAGBoost and competing tabular models under five synthetic data scales. The reported values are mean ± standard deviation over repeated runs. The main text visualizes these trends using line plots, while the full numerical results are provided here for completeness.

Table C.1: Training runtime (seconds) across data scales. Mean ± std over repeated runs.

Method	$(1k\times10)$	$(5k\times30)$	$(10k\times60)$	$(50k\times90)$	$(100k\times120)$
XGBoost	0.290±0.073	0.510±0.016	1.315±0.040	3.094±0.104	8.439±0.087
CatBoost	0.497±0.093	0.894±0.015	1.820±0.061	4.005±0.061	8.910±0.009
LightGBM	0.049±0.004	0.193±0.003	0.423±0.067	1.841±0.019	3.562±0.056
RAGBoost(rag)	0.155±0.008	0.591±0.002	1.564±0.034	5.205±0.014	9.003±0.077
RAGBoost(rag+anc)	0.311±0.007	1.795±0.019	4.695±0.066	16.171±0.068	30.515±0.473
MLP	0.436±0.348	0.746±0.016	1.455±0.032	10.648±0.377	17.663±0.226
TabM	0.739±0.057	3.812±0.037	9.678±0.058	58.452±0.302	150.000±2.556
TabPFN	0.783±0.577	2.672±0.014	14.144±0.069	301.556±2.302	Exceeds limit

The empirical results are consistent with the theoretical complexity decomposition. RAGBoost(rag), which activates only the retrieval-augmented shrinkage layer, introduces a moderate overhead over standard tree boosting. At the largest scale, (100000×120), RAGBoost(rag) requires 9.003 ± 0.077 seconds, which is close to XGBoost (8.439 ± 0.087 seconds) and CatBoost (8.910 ± 0.009 seconds), although slower than LightGBM (3.562 ± 0.056 seconds). This supports the interpretation that the retrieval term is practically manageable when K, q, and the admitted-leaf set are controlled.

RAGBoost(rag+anc) is consistently slower than RAGBoost(rag), which matches the additional $TPC_{\text{anc}}(n, d) + TPn$ terms introduced by the ancillary correction layer. At (100000×120), the full model requires 30.515 ± 0.473 seconds. This increase is expected because ancillary trees are fitted at every boosting iteration and diagnostic quantities are accumulated over samples. However, the runtime remains below several larger neural or foundation-style tabular baselines at high data scales. For example, at (50000×90), RAGBoost(rag+anc) takes 16.171 ± 0.068 seconds, while TabM takes 58.452 ± 0.302 seconds and TabPFN takes 301.556 ± 2.302 seconds. At (100000×120), TabPFN exceeds its model limit, whereas RAGBoost remains trainable.

C.3 Summary

The theoretical and empirical results lead to the same conclusion. RAGBoost increases training-time complexity mainly through two controlled mechanisms: memory retrieval in Layer 1 and transient ancillary-tree fitting in Layer 2. The retrieval-only variant adds a relatively small overhead and remains close to conventional boosting baselines at large scale. The full variant is more expensive because of ancillary diagnostics, but the increase follows the expected complexity terms and remains practically manageable in the tested regimes.

More importantly, the extra cost is limited to training. Since neither the memory bank nor the ancillary trees are used at inference time, RAGBoost preserves the deployment complexity of standard GBDT while improving leaf-weight estimation during training. This analysis highlights an important design property of RAGBoost: its statistical correction mechanisms are decoupled from deployment complexity. RAGBoost therefore improves training-time estimation through controlled optimization-side computation, rather than by enlarging the inference-time model.

D Experimental Details

All experiments are conducted under the same hardware and software environment. The machine is equipped with an Intel Core i7-12700KF CPU, an NVIDIA RTX A4000 GPU, and consistent RAM, operating system, Python environment, and package versions for all compared methods. For each dataset and method, we use the same data-loading pipeline, random seed sequence, train/validation/test split, and hyperparameter optimization budget.

D.1 Public Benchmark Datasets

We evaluate RAGBoost on 40 public tabular datasets, including 20 classification and 20 regression tasks. The datasets are selected to cover different sample sizes, feature dimensions, feature types, and target distributions. Tables D.1 and D.2 summarize the dataset metadata. For all public datasets, we apply a unified preprocessing pipeline across all methods. Missing values are removed or imputed using statistics computed from the training split only. Categorical variables are encoded into numerical representations using a consistent encoding scheme, and all resulting features are treated as continuous inputs. Numerical features are normalized using training-set statistics and the same transformation is applied to validation and test splits.

D.2 TabReD Benchmark

We further evaluate RAGBoost on TabReD, an industrial tabular benchmark designed to reflect practical deployment conditions. Unlike conventional academic benchmarks that commonly rely on random splits, TabReD emphasizes temporally evolving data distributions and feature-rich real-world datasets. We strictly follow the official TabReD protocol, including the predefined time-based splits, task-specific metrics, and evaluation pipeline.

D.3 Preprocessing and Splitting

For the 40 public datasets, all methods are evaluated on exactly the same train/validation/test splits. Each dataset is split into 70% training, 15% validation, and 15% test data. For classification tasks, stratified splitting is used whenever applicable. For regression tasks, random splitting is performed using the same seed sequence across all methods. The validation set is used only for hyperparameter

Table D.1: Metadata of the 20 public classification datasets.

Dataset	Samples	Features	Classes	Shape
adult	48842	14	2	(48842, 15)
autos	205	25	6	(205, 26)
bank_marketing	45211	16	2	(45211, 17)
breast_w	699	9	2	(699, 10)
car	1728	6	4	(1728, 7)
credit_g	1000	20	2	(1000, 21)
diabetes	768	8	2	(768, 9)
dna	3186	180	3	(3186, 181)
eeg_eye	14980	14	2	(14980, 15)
glass	214	9	6	(214, 10)
har	10299	561	6	(10299, 562)
iris	150	4	3	(150, 5)
madelon	2600	500	2	(2600, 501)
magic_telescope	19020	10	2	(19020, 11)
musk	6598	166	2	(6598, 167)
pendigits	10992	16	10	(10992, 17)
phoneme	5404	5	2	(5404, 6)
satimage	6430	36	6	(6430, 37)
vehicle	846	18	4	(846, 19)
wine	178	13	3	(178, 14)

Table D.2: Metadata of the 20 public regression datasets.

Dataset	Samples	Features	Shape
airfoil	1503	5	(1503, 6)
beijing_pm25	41757	12	(41757, 13)
bike_sharing	17379	12	(17379, 13)
california_housing	20640	8	(20640, 9)
casp	45730	9	(45730, 10)
ccpp_dataset	9568	4	(9568, 5)
concrete	1030	8	(1030, 9)
cpu_active	8192	21	(8192, 22)
pol	15000	26	(15000, 27)
elevators	16599	18	(16599, 19)
enb2012	768	8	(768, 9)
energy	768	8	(768, 9)
kin8nm	8192	8	(8192, 9)
meps	15675	139	(15675, 140)
naval_dataset	11934	16	(11934, 17)
news	39644	60	(39644, 61)
sensory	576	11	(576, 12)
superconduct	21263	81	(21263, 82)
wine_red	1599	11	(1599, 12)
yacht_hydrodynamics	308	6	(308, 7)

selection, and the test set is used only for final reporting. For TabReD, we use the official temporal train/validation/test splits and the official evaluation pipeline.

D.4 Hyperparameter Optimization Protocol

For the 40 public datasets, all tunable methods are optimized using Bayesian optimization with the same budget of 50 trials per dataset and per random seed. Each candidate configuration is trained on the training split and selected according to validation performance. The same split, seed sequence, optimization budget, and metric direction are used for all methods.

Table D.3: Metadata of the TabReD datasets. "Used" denotes the number of instances used in the benchmark protocol, and "Available" denotes the total number of available records when provided by TabReD.

Dataset	# Features	Task	Used	Available
Homesite Insurance	299	Classification	260,753	–
Ecom Offers	119	Classification	160,057	–
HomeCredit Default	696	Classification	381,664	1,526,659
Sberbank Housing	392	Regression	28,321	–
Cooking Time	192	Regression	319,986	12,799,642
Delivery ETA	223	Regression	416,451	17,044,043
Maps Routing	986	Regression	340,981	13,639,272
Weather	103	Regression	423,795	16,951,828

After hyperparameter selection, the selected configuration is evaluated once on the held-out test split. Each experiment is repeated over 10 random seeds, and we report the mean performance across seeds. This protocol ensures that the reported differences are not caused by different data splits, unequal tuning budgets, or method-specific access to the test set.

TabPFN is evaluated using its default pretrained inference procedure without additional hyperparameter tuning. For TabReD, we follow the official benchmark setup and use the official validation and test splits.

D.5 Hyperparameter Search Spaces

All tunable baselines are optimized under the same 50-trial Bayesian optimization budget. The search spaces are summarized in Tables D.4 through D.6. TabPFN is used without additional HPO.

D.6 Metrics and Aggregation

For public classification datasets, we use Accuracy. For public regression datasets, we use RMSE. To summarize performance across datasets, we report average rank, average relative score, and the number of dataset-level wins.

For each dataset, methods are ranked according to the corresponding task metric, with higher Accuracy and lower RMSE considered better. The average rank is computed by averaging dataset-level ranks across the corresponding benchmark group. The number of wins counts the number of datasets on which a method obtains the best test performance.

The average relative score is computed independently for each dataset. For classification, where higher Accuracy is better, the relative score of method m on dataset d is:

$$s_{m,d} = \frac{a_{m,d} - \min_{m'} a_{m',d}}{\max_{m'} a_{m',d} - \min_{m'} a_{m',d} + \epsilon},$$

where $a_{m,d}$ denotes Accuracy. For regression, where lower RMSE is better, we use reversed min-max normalization:

$$s_{m,d} = \frac{\max_{m'} e_{m',d} - e_{m,d}}{\max_{m'} e_{m',d} - \min_{m'} e_{m',d} + \epsilon},$$

where $e_{m,d}$ denotes RMSE. We use a small numerical constant ϵ to avoid division by zero. The final average relative score is obtained by averaging $s_{m,d}$ across datasets.

For TabReD, we follow the official task metrics: ROC AUC for classification and RMSE for regression. We additionally report average rank across the eight TabReD tasks.

D.7 Full Experiment Results

Table D.7 reports the full classification results on the 20 public datasets. Table D.8 reports the full regression results. Table D.9 reports the full results on TabReD.

Table D.4: Search spaces for tree-based methods.

Method	Hyperparameter	Search Space
XGBoost	n_estimators	{500, ..., 3000}
	max_depth	{3, ..., 12}
	learning_rate	[0.01, 0.3] log-uniform
	min_child_weight	{1, ..., 10}
	gamma	[10^{-6}, 100] log-uniform
	reg_lambda	[0.1, 10] log-uniform
	subsample	[0.5, 1.0]
	colsample_bytree	[0.5, 1.0]
LightGBM	n_estimators	{500, ..., 3000}
	num_leaves	{16, ..., 256}
	max_depth	{-1, 3, ..., 12}
	learning_rate	[0.01, 0.3] log-uniform
	min_child_samples	{5, ..., 100}
	reg_lambda	[0.1, 10] log-uniform
	subsample	[0.5, 1.0]
	colsample_bytree	[0.5, 1.0]
CatBoost	iterations	{500, ..., 3000}
	depth	{3, ..., 10}
	learning_rate	[0.01, 0.3] log-uniform
	l2_leaf_reg	[0.1, 10] log-uniform
	random_strength	[10^{-3}, 10] log-uniform
	bagging_temperature	[0, 1]
RandomForest	n_estimators	{500, ..., 3000}
	max_depth	{3, ..., 30} $\cup$ {None}
	max_features	[0.3, 1.0]
	min_samples_split	{2, ..., 20}
	min_samples_leaf	{1, ..., 10}

D.8 Statistical Significance Analysis

To further evaluate the consistency and statistical significance of RAGBoost's performance, we conduct nonparametric statistical analyses based on the experimental results of 40 public datasets, comprising 20 classification and 20 regression tasks, as reported in Tables D.7 and D.8. The analysis includes the Friedman test for overall method comparison, the Nemenyi post-hoc test for identifying significant differences between methods, and the Wilcoxon signed-rank test for paired performance comparisons. We additionally examine the results on TabReD to assess performance under temporal distribution shift.

D.8.1 Friedman Test: Overall Performance Comparison

We first employ the Friedman test to determine whether the nine methods exhibit statistically significant performance differences across the 40 public datasets. For each dataset, methods are ranked according to their predictive performance, with higher Accuracy preferred for classification and lower RMSE preferred for regression. Tied performances receive average ranks.

The average rank of method j across N datasets is calculated as

$$\bar{R}_j = \frac{1}{N} \sum_{i=1}^{N} r_{ij},$$

where r_{ij} denotes the rank of method j on dataset i. The Friedman statistic is defined as

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^{k} \bar{R}_j^2 - \frac{k(k+1)^2}{4} \right],$$

where $N = 40$ and $k = 9$. The standard tie correction is applied when calculating the test statistic.

Table D.5: Search spaces for neural, retrieval-based, and recent tabular baselines.

Method	Hyperparameter	Search Space
MLP	hidden dimension	{64, ..., 512}
	number of layers	{2, ..., 5}
	learning rate	$[10^{-4}, 10^{-1}]$ log-uniform
	batch size	{32, ..., 1024}
	dropout	[0.0, 0.5]
	weight decay	[0.0, 0.1]
FT-Transformer	token dimension	{64, ..., 512}
	number of layers	{2, ..., 6}
	number of heads	{4, 8}
	learning rate	$[10^{-4}, 10^{-1}]$ log-uniform
	attention dropout	[0.0, 0.3]
	residual dropout	[0.0, 0.3]
	weight decay	[0.0, 0.1]
TabR	hidden dimension	{64, ..., 512}
	number of blocks	{2, ..., 5}
	context size	{16, 32, 64}
	learning rate	$[10^{-4}, 10^{-1}]$ log-uniform
	batch size	{32, ..., 1024}
	dropout	[0.0, 0.5]
	weight decay	[0.0, 0.1]
TabM	d_block	{64, ..., 512}
	n_blocks	{2, ..., 5}
	learning rate	$[10^{-4}, 10^{-1}]$ log-uniform
	k	32 (fixed)
	patience	16 (fixed)
	batch size	{32, ..., 1024}
	weight decay	[0.0, 0.1]
TabPFN	–	default pretrained inference

The results indicate that RAGBoost achieves the lowest overall average rank of 2.9750, compared with 4.0125 for TabPFN, 4.4000 for TabM, 5.1625 for XGBoost, 5.8125 for LightGBM, and 6.5875 for RandomForest. When evaluated separately, RAGBoost also achieves the most favorable average ranks on both classification (3.2250) and regression (2.7250) tasks. It obtains the top or tied-top performance on 13 of the 40 datasets and ranks among the top three on 27 datasets.

The Friedman test produces a tie-adjusted statistic of $\chi_F^2 = 48.43$, with 8 degrees of freedom and $p = 8.16 \times 10^{-8}$. This result demonstrates statistically significant performance differences among the evaluated methods. Together with its lowest overall average rank, this suggests RAGBoost's robust aggregate performance across diverse classification and regression benchmarks.

D.8.2 Nemenyi Post-hoc Test: Pairwise Performance Differences

Following the Friedman test, we apply the Nemenyi post-hoc test to identify which pairwise differences are statistically significant. At a family-wise significance level of $\alpha = 0.05$, two methods are considered significantly different when their average-rank difference exceeds the critical difference (CD), calculated as

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}},$$

where q_α is the corresponding critical value of the studentized-range statistic under the Nemenyi convention.

For $k = 9$ methods and $N = 40$ datasets, the resulting critical difference is $CD = 1.8994$.

Table D.6: Additional search space for RAGBoost.

Component	Hyperparameter	Search Space
GBDT	n_estimators	{500, ..., 3000}
GBDT	max_depth	{3, ..., 12}
GBDT	learning_rate	[0.01, 0.3] log-uniform
GBDT	min_child_weight	{1, ..., 10}
Retrieval	c_{rag}	[0.1, 5.0]
Retrieval	γ_{rag}	[0.1, 1.0]
Retrieval	ρ_{rag}	[0.1, 1.0]
Retrieval	K	{3, ..., 30}
Retrieval	warmup rounds	{5, ..., 50}
Retrieval	memory capacity	{1000, ..., 3000}
Ancillary	enabled	{True, False}
Ancillary	column sampling ratio	[0.2, 0.8]
Ancillary	η_{anc}	[0.05, 0.5]
Ancillary	τ_E	[0.05, 0.5]
Ancillary	λ_{anc}	[0.05, 1.0]
Ancillary	$s_{\max}$	[1.5, 5.0]
Ancillary	$\varepsilon_{\text{gain}}$	[0, 0.01]

Table D.7: Full classification results on the 20 public datasets. Accuracy is reported as mean $\pm$ standard deviation over 10 random seeds. Higher is better.

Dataset	RAGBoost	XGBoost	LightGBM	CatBoost	RF	FT-Trans	TabR	TabM	TabPFN
adult	0.8795±0.0030	0.8761±0.0014	0.8749±0.0002	0.8649±0.0021	0.8607±0.0003	0.8728±0.0047	0.8758±0.0011	0.8654±0.0016	0.8037±0.0012
autos	0.8750±0.0003	0.8350±0.0443	0.8732±0.0055	0.8750±0.0062	0.7957±0.0203	0.8749±0.0066	0.8350±0.0443	0.8056±0.0241	0.8893±0.0000
bank_marketing	0.9157±0.0064	0.9023±0.0034	0.9028±0.0015	0.8989±0.0031	0.9087±0.0029	0.8917±0.0023	0.9028±0.0085	0.8991±0.0010	0.8984±0.0005
breast_w	0.9517±0.0041	0.9417±0.0097	0.9515±0.0015	0.9392±0.0016	0.9509±0.0056	0.9405±0.0040	0.9425±0.0037	0.9420±0.0033	0.9476±0.0012
car	0.9899±0.0033	0.9885±0.0102	0.9808±0.0038	0.9897±0.0146	0.9423±0.0102	0.9650±0.0200	0.9975±0.0010	1.0000±0.0000	0.9563±0.0013
credit_g	0.7587±0.0410	0.7400±0.0067	0.7533±0.0133	0.7378±0.0214	0.7378±0.0139	0.7350±0.0250	0.7330±0.0800	0.7433±0.0236	0.7337±0.0000
diabetes	0.7959±0.0013	0.7989±0.0100	0.7931±0.0086	0.7874±0.0100	0.7960±0.0132	0.8050±0.0250	0.7200±0.1500	0.7954±0.0061	0.7601±0.0000
dna	0.9544±0.0005	0.9644±0.0055	0.9693±0.0044	0.9596±0.0053	0.9317±0.0032	0.9522±0.0030	0.9519±0.0075	0.9623±0.0030	0.9634±0.0000
eeg_eye	0.9507±0.0027	0.9464±0.0053	0.9387±0.0032	0.9504±0.0053	0.9113±0.0027	0.9663±0.0105	0.9722±0.0098	0.9722±0.0009	0.9847±0.0000
glass	0.6761±0.0178	0.6465±0.0350	0.5857±0.0463	0.6364±0.0000	0.6464±0.0310	0.9050±0.0150	0.9102±0.0250	0.6212±0.0214	0.4868±0.0022
har	0.9881±0.0024	0.9855±0.0040	0.9841±0.0005	0.9860±0.0009	0.9852±0.0024	0.9950±0.0050	0.9990±0.0008	0.9899±0.0004	0.9987±0.0000
iris	1.0000±0.0000	0.9130±0.0000	0.9710±0.0251	0.9565±0.0000	0.9565±0.0000	1.0000±0.0000	0.9712±0.0251	0.9565±0.0000	1.0000±0.0000
madelon	0.8392±0.0000	0.8325±0.0126	0.8137±0.0299	0.8267±0.0299	0.8376±0.0030	0.8150±0.0180	0.8190±0.0170	0.8308±0.0218	0.9051±0.0066
MagicTelescope	0.8805±0.0027	0.8800±0.0019	0.8774±0.0007	0.8791±0.0026	0.8668±0.0015	0.8633±0.0078	0.8655±0.0015	0.8784±0.0030	0.8780±0.0000
musk	0.9869±0.0000	0.9852±0.0021	0.9865±0.0015	0.9879±0.0010	0.9795±0.0012	0.9750±0.0060	0.9758±0.0007	0.9924±0.0007	0.9981±0.0003
pendigits	0.9969±0.0000	0.9915±0.0006	0.9913±0.0009	0.9947±0.0004	0.9852±0.0091	0.9985±0.0010	0.9993±0.0003	0.9961±0.0004	0.9988±0.0002
phoneme	0.8545±0.0000	0.8627±0.0071	0.8598±0.0031	0.8726±0.0061	0.8664±0.0026	0.8469±0.0036	0.9505±0.0120	0.8767±0.0052	0.8532±0.0000
satimage	0.9297±0.0022	0.9264±0.0055	0.9264±0.0021	0.9275±0.0036	0.9275±0.0010	0.9295±0.0030	0.9185±0.0073	0.9337±0.0015	0.9211±0.0001
vehicle	0.7638±0.0000	0.8058±0.0045	0.7709±0.0192	0.7822±0.0164	0.7795±0.0236	0.7612±0.0198	0.7717±0.0284	0.8228±0.0056	0.8728±0.0000
wine	0.9632±0.0008	0.9630±0.0022	0.9630±0.0002	0.9630±0.0000	0.9630±0.0000	0.9510±0.0015	0.9586±0.0020	0.9444±0.0262	0.9666±0.0055

The average-rank differences between RAGBoost and XGBoost (2.1875), LightGBM (2.8375), CatBoost (2.3000), RandomForest (3.6125), FT-Transformer (2.4250), and TabR (2.4000) all exceed this threshold.

Therefore, RAGBoost exhibits statistically significant improvements over six of the eight competing methods under the Nemenyi test. These results provide statistical evidence that its performance advantage is consistent across both classical tree-based methods and deep tabular learning approaches.

D.8.3 Wilcoxon Signed-Rank Test: Paired Comparison with XGBoost

We further conduct two-sided Wilcoxon signed-rank tests to examine the dataset-level performance differences between RAGBoost and XGBoost, a widely adopted gradient-boosting baseline. The test is applied separately to the paired Accuracy values of the 20 classification datasets and the paired RMSE values of the 20 regression datasets.

On the classification benchmarks, RAGBoost achieves higher Accuracy than XGBoost on 16 of the 20 datasets. The Wilcoxon signed-rank test yields $p = 0.0419$, indicating a statistically significant improvement at the 0.05 level.

Table D.8: Full regression results on the 20 public datasets. RMSE is reported as mean ± standard deviation over 10 random seeds. Lower is better.

Dataset	RAGBoost	XGBoost	LightGBM	CatBoost	RF	FT-Trans	TabR	TabM	TabPFN
airfoil	1.2630±0.0042	1.4134±0.0665	1.3106±0.0506	1.3155±0.0235	1.1405±0.0939	1.3450±0.0280	1.3200±0.0350	1.3332±0.0931	1.0166±0.0125
beijing_pm25	27.1958±0.6989	29.3275±0.5313	30.6382±0.1418	31.3579±2.3848	32.0781±1.1456	30.2000±1.5000	31.9000±2.1000	27.1960±1.2283	27.3820±1.0404
bike_sharing	35.2854±0.0484	35.6016±1.0772	37.8163±1.1116	39.0889±1.5457	39.7331±1.8931	34.9000±1.2000	33.1000±1.8000	30.3515±1.0254	45.4760±2.1041
california_housing	0.4307±0.0065	0.4408±0.0030	0.4334±0.0012	0.4369±0.0047	0.4951±0.0145	0.4350±0.0100	0.4500±0.0180	0.4318±0.0054	0.4017±0.0000
casp	3.4012±0.0120	3.4592±0.0509	3.5040±0.0311	3.6234±0.0768	3.6230±0.0681	3.1200±0.0600	3.6500±0.0900	3.3691±0.0186	2.4153±0.6575
ccpp_dataset	0.2757±0.0023	0.3761±0.0442	0.5195±0.0918	0.5820±0.0651	0.6104±0.2812	0.6605±0.0315	0.4500±0.1200	0.5586±0.0180	0.6198±0.0013
concrete	4.1054±0.0025	4.0083±0.1186	4.2133±0.0301	4.2671±0.0990	4.3506±0.3016	4.2050±0.0298	4.3008±0.6000	3.8629±0.0203	4.4613±0.0009
cpu_active	2.1501±0.0000	2.2848±0.0082	2.3067±0.0232	2.3234±0.0104	2.3903±0.0622	2.2700±0.0030	2.2805±0.0020	2.3395±0.0000	2.1772±0.0000
pol	4.2902±0.1263	4.5336±0.0963	4.3322±0.0963	4.3122±0.0963	4.5346±0.0783	4.5223±0.0669	4.4336±0.0963	2.2071±0.0134	2.5729±0.0629
elevators	0.0000±0.0000	0.0025±0.0001	0.0031±0.0003	0.0002±0.0001	0.0022±0.0000	0.0018±0.0002	0.0028±0.0001	0.0003±0.0000	0.0000±0.0000
enb2012	0.3798±0.0000	0.4017±0.0478	0.4652±0.0637	0.3429±0.0264	0.4003±0.0095	0.4172±0.0455	0.4223±0.0381	0.5844±0.0000	0.4463±0.0000
energy	0.8232±0.0000	1.0472±0.0845	1.1718±0.0979	0.8603±0.0053	1.4494±0.1333	0.8600±0.0050	1.0502±0.0600	1.1533±0.0094	1.0785±0.0041
kin8nm	0.1004±0.0010	0.1057±0.0010	0.1161±0.0013	0.0874±0.0017	0.1493±0.0020	0.0670±0.0015	0.0760±0.0023	0.0663±0.0011	0.0660±0.0020
meps	0.3462±0.0177	0.3685±0.0379	0.3540±0.0201	0.3466±0.0127	0.3584±0.0593	0.3350±0.0300	0.3650±0.0200	0.3482±0.0098	0.3396±0.0070
naval_dataset	0.0003±0.0000	0.0032±0.0001	0.0007±0.0000	0.0005±0.0001	0.0008±0.0002	0.00035±0.00005	0.00045±0.00005	0.0003±0.0000	0.0001±0.0000
news	11873.1205±41.1209	11890.7059±35.0980	11879.1653±55.3141	11913.5683±42.7112	11899.7155±44.7661	11932.7059±45.7550	11890.7059±34.7550	11912.7005±46.8178	11875.7005±45.8258
sensory	0.7126±0.0054	0.7244±0.0104	0.7334±0.0088	0.7251±0.0241	0.7364±0.0051	0.7244±0.0106	0.7244±0.0107	0.7306±0.0047	0.7221±0.0025
Superconduct	9.4670±0.9885	9.2261±0.0492	9.4566±0.0408	9.5224±0.0523	9.4116±0.0588	9.4428±0.0775	9.3044±0.0322	9.5720±1.0032	9.2228±0.3975
wine_red	0.5795±0.0023	0.5795±0.0071	0.5792±0.0133	0.5489±0.0145	0.5841±0.0406	0.5850±0.0300	0.6050±0.0300	0.5639±0.0067	0.6199±0.0000
yacht_hydrodynamics	0.7293±0.0903	0.7609±0.1771	0.7824±0.1952	0.7535±0.1331	0.7649±0.8122	0.7688±0.0105	0.7559±0.1200	0.5962±0.1022	0.7209±0.0000

Table D.9: Full results on TabReD. ROC AUC is used for classification tasks and RMSE for regression tasks. Best, second-best, and third-best results are highlighted in **bold**, *italic*, and underline, respectively.

Methods	Classification (ROC AUC ↑)			Regression (RMSE ↓)					Avg. Rank
	Homesite	Ecom	Credit	Sberbank	Cooking	Delivery	Maps	Weather	
RAGBoost(ours)	0.9608	0.5844	**0.8677**	0.2440	0.4823	**0.5465**	0.1622	**1.4624**	*3.56*
XGBoost	0.9601	0.5763	*0.8670*	*0.2419*	0.4823	0.5468	0.1616	1.4671	4.44
LightGBM	0.9603	0.5758	0.8664	0.2468	0.4826	0.5468	0.1618	*1.4625*	5.19
CatBoost	0.9606	0.5596	0.8621	0.2482	0.4823	**0.5465**	0.1619	1.4688	5.69
RandomForest	0.9570	0.5764	0.8269	0.2640	0.4884	0.5959	0.1653	1.5838	9.25
MLP	0.9500	**0.6015**	0.8545	0.2508	0.4820	0.5504	0.1622	1.5470	6.38
FT-Transformer	0.9622	0.5775	0.8571	0.2440	0.4820	0.5542	0.1625	1.5104	6.00
TabR	0.9487	0.5943	0.8501	0.2820	0.4828	0.5514	0.1639	1.4666	7.50
TabM	**0.9647**	*0.5950*	0.8614	0.2467	**0.4802**	0.5502	*0.1612*	1.4706	3.69
$\text{TabM}^{\dagger}_{\text{mini}}$	*0.9631*	0.5930	0.8640	**0.2319**	**0.4802**	0.5516	**0.1609**	1.4672	**3.31**

On the regression benchmarks, RAGBoost achieves lower RMSE on 17 of the 20 datasets, with one additional dataset showing equal reported performance. The corresponding Wilcoxon test produces $p = 0.00485$, indicating a statistically significant improvement in regression performance.

We additionally perform a joint paired analysis across all 40 datasets using within-dataset rank differences, providing a unified comparison across classification and regression tasks. The two-sided Wilcoxon test yields $p = 1.49 \times 10^{-4}$, further supporting the performance advantage of RAGBoost over XGBoost.

These findings indicate that RAGBoost's improvements are consistently observed at the individual-dataset level and remain statistically significant across both predictive task types.

D.8.4 Performance Analysis on TabReD

The eight-task TabReD benchmark is assessed separately from the 40 public datasets. RAGBoost has an average rank of 3.56, compared with 4.44 for XGBoost, 5.19 for LightGBM, 5.69 for CatBoost, and 3.69 for TabM. Given the small number of tasks, these rank differences should be interpreted cautiously. In a two-sided paired Wilcoxon analysis of per-task ranks, no RAGBoost-versus-baseline comparison remains significant at the 5% level after Holm correction across the nine comparisons. Thus, TabReD supports the claim of competitive performance under temporal distribution shift, rather than statistically established superiority over individual baselines.

E Analysis Details

This appendix describes the five DAG-generated regression benchmarks used in Section 5. The construction follows the structural causal data generation perspective used in recent tabular foundation model research: a dataset is sampled from an underlying structural mechanism consisting of latent noise variables, observed feature nodes, intermediate computation nodes, and a target node. By varying the structural equations, nonlinear transformations, interaction patterns, and noise mechanisms, we obtain datasets that expose different local estimation regimes of gradient-boosted trees.

Each benchmark contains 50,000 training samples, 10,000 validation samples, and 10,000 test samples. All tasks use $d = 30$ observed features and a scalar regression target. The evaluation metric is RMSE. For random-split benchmarks, train, validation, and test sets are obtained by random partitioning. For the temporal-shift benchmark, the split is ordered by the time variable.

E.1 General Structural Layer

For each benchmark, we first sample latent variables:

$$Z \sim \mathcal{N}(0, I_m)$$

and generate observed covariates through:

$$X = ZB + \epsilon_X,$$

where $B \in \mathbb{R}^{m \times d}$ is a randomly sampled loading matrix and ϵ_X is independent Gaussian noise. The observed features are standardized column-wise. The response is generated by:

$$y = f(X, Z, T) + \epsilon_y,$$

where the structural function f differs across benchmarks.

E.2 DAG-Sparse

DAG-Sparse generates low-probability local regions with strong effects. Define:

$$S = \mathbb{1}(X_1 > 1.7,\ X_2 > 1.2).$$

The response is:

$$y = 2X_1 - 1.5X_2 + \sin(X_3) + 4S + (1 + 2S)\epsilon,$$

where $\epsilon \sim \mathcal{N}(0, 1)$. The rare event $S = 1$ creates a local region with both stronger signal and larger noise. Tree learners tend to represent this region using small leaves, which makes the corresponding leaf updates sensitive to sampling variation.

E.3 DAG-Hetero

DAG-Hetero introduces interaction-induced residual heterogeneity. Define:

$$G = \mathbb{1}(X_4 X_5 > 0.8).$$

The response is:

$$y = 1.5X_1 + X_2^2 - X_3 + 3G\sin(2X_6) + \epsilon.$$

The interaction $X_4 X_5$ determines whether X_6 becomes active in the target mechanism. If the main tree does not isolate this interaction cleanly, samples with different residual directions may be routed into the same leaf.

E.4 DAG-Repeated

DAG-Repeated creates several recurring local mechanisms. Let:

$$C = \arg\max_{k \in \{1,2,3,4\}} (a_k^\top X + \xi_k),$$

where a_k are randomly sampled regime vectors and ξ_k are independent noise variables. The base function is:

$$f_0(X) = 1.2\sin(X_1) + 0.8X_2 - 0.5X_3^2.$$

The response is:

$$y = f_0(X) + \sum_{k=1}^{4} \mathbb{1}(C = k)\, b_k^\top X_{S_k} + \epsilon.$$

Here S_k denotes a regime-specific feature subset. Although the regimes occupy different feature regions, their local gradient-state patterns can be partially shared. This benchmark is therefore useful for analyzing retrieval provenance and memory consistency.

E.5 DAG-Shift

DAG-Shift introduces a time variable:

$$T \sim \mathrm{Uniform}(0, 1).$$

The time-dependent observed features are:

$$X_1 = Z_1 + 0.8T, \qquad X_2 = Z_2 - 0.5T.$$

The response is:

$$y = 2X_1 + X_2 + \sin(X_3 + T) + 2.5\mathbb{1}(X_4 + T > 1.2)\, X_5 + \epsilon.$$

Samples are split according to T, with earlier samples used for training and later samples used for testing. This benchmark creates both covariate drift and mild mechanism drift.

E.6 DAG-NoisyRare

DAG-NoisyRare creates a rare subgroup dominated by high irreducible noise. Define:

$$R = \mathbb{1}(X_1 < -1.5,\ X_2 > 1.5).$$

The base function is:

$$f_0(X) = 1.5X_1 + X_2 + 0.7\sin(X_3) + 0.5X_4X_5.$$

The response is:

$$y = f_0(X) + (1 - R)\epsilon + R\epsilon_R,$$

where $\epsilon \sim \mathcal{N}(0, 1)$ and $\epsilon_R \sim \mathcal{N}(0, 6^2)$. This setting separates rarity from reliability. The rare subgroup is statistically difficult not because it contains a stable rare signal, but because it is dominated by noise. A reliable correction mechanism should therefore be conservative in this region.

E.7 Recorded Diagnostics

For each trained RAGBoost model, we record both leaf-level and sample-level diagnostics.

Leaf-level diagnostics. Include leaf sample count, residual variance, standard Newton update, corrected update, correction magnitude, retrieval admission indicator, retrieval gate value, retrieval uncertainty, ancillary alignment score, diagnostic reweighting score, and correction state.

Sample-level diagnostics. Include prediction decomposition, visited correction states, retrieval-neighbor provenance, ancillary contribution, and SHAP values.

Correction states are grouped into standard update, retrieval correction, ancillary correction, diagnostic reweighting, and fallback. We report sample-weighted state distributions because they better reflect prediction-time exposure than raw leaf counts.

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   Justification: The abstract claims that RAGBoost addresses the variance-bias tension in deep GBDT leaves via a two-layer node-adaptive correction mechanism (retrieval-augmented empirical shrinkage and ancillary residual correction). The paper thoroughly supports this through formal theoretical analysis (Theorems 3.1–3.2, Propositions 3.1–3.2 in Sections 3 and Appendix A) and extensive empirical validation across 40 public tabular benchmarks and the industrial-scale TabReD benchmark (Section 4).

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- Methodology: The core RAGBoost algorithm is fully specified in Algorithm 1 and Appendix A, including all gating functions, admission criteria, and Layer 2 diagnostic reweighting details.
- Datasets & Preprocessing: All 40 public benchmark datasets are listed with their sources (OpenML/Kaggle) and metadata (Tables D.1–D.2). The preprocessing pipeline, data splits (70/15/15), and TabReD's official protocol are explicitly described in Appendix D.3.
- Hyperparameter Search: A unified Bayesian optimization protocol with an equal budget of 50 trials is used for all methods. The complete search spaces for all baselines and RAGBoost are detailed in Appendix D.5 (Tables D.4–D.6). Hyperparameters were tuned on a validation set, and final performance is reported on a held-out test set over 10 random seeds.
- Code Release: We are committed to releasing the complete source code, including the core RAGBoost implementation, a packaged Python library (`ragboost`), all experiment scripts, and the integration scripts for the TabReD benchmark. The code will be made publicly available under an open-source license upon publication.

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